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1.0 DESCRIPTION OF WORK – GENERAL

Except as otherwise expressly provided herein, Contractor shall supply all adequate and competent labor, supervision, spare parts, tools, equipment, installed and consumable materials, services, testing devices and warehousing and each and every item of expense necessary for the supply of Rolling Stock which includes the design, manufacture, assembly, delivery to site, testing and commissioning, support for the commencement of revenue service and maintenance training for new Electric Multiple Unit (EMU) commuter rail passenger Cars, hereinafter called the Work or Works. Company's base order for the East Corridor, Gold Line and Northwest Electrified Segment to be 50 individual Cars with Company Rolling Stock Option to purchase between 2 and 10 additional Cars and Owner's Rolling Stock Option to purchase an additional 8 to 24 additional Cars. The Cars shall be suitable in all respects for commuter rail operation at speed limits up to 79 mph, on both dedicated track and track shared with freight service. Rolling Stock must comply with Federal Railroad Administration (FRA) requirements for commuter rail operations on a shared Class 1 Railway freight alignment and requirements for passenger cars present in 49 CFR.

2.0 SPECIFICATIONS, DRAWINGS, ATTACHMENTS AND EXHIBITS

All Work shall be performed in strict accordance with the following described specifications, drawings and other documents, which by this reference are made a part hereof.

2.1 Specifications

Specification No.	Date	Title
Contract No. 18FH012 Concession Agreement	9 July 2010	Attachment 6 to the Concession Agreement, Contract Data Requirements List (CDRL)
Contract No. 18FH012 Concession Agreement	9 July 2010	Attachment 7, Design, Construction and Rolling Stock Requirements
Alternate Technical Concepts (ATCs) accepted by RTD		DTP-ATC-0011 HVAC, Lighting, Equipment Configuration/Redundancy and Side Doors DTP-ATC-0023 Passenger Seat Armrest Configuration DTP-ATC-0024 Rolling Stock Thermal Insulation DTP-ATC-0027 Rolling Stock Optimisation
Contract No. 18FH012 Concession Agreement	9 July 2010	Attachment 10 to the Concession Agreement, O&M Specifications
Contract No. 18FH012 Concession Agreement	9 July 2010	Attachment 9 to the Concession Agreement, Project and Construction Management

Contract No. 18FH012	9 July 2010	Attachment 19 to the Concession Agreement, Concessionaire's Proposal
Concession Agreement 2RS	10 Aug 2009	Approach to Achieving Level Boarding for the Commuter Rail System
DTP Proposal Volume 3 C.12		Technical Approach Rolling Stock Sections A-S Technical Description of the Rolling Stock Design Concept and Technical Approach
4RR	Dec 2008	Steep Grade Preliminary Hazard Analysis
A159-TP-D112	Proposal Revision	Hyundai Rotem Drawing - Static Envelope
A159-TP-D-113	Proposal Revision	Hyundai Rotem Drawing - Dynamic envelope

2.2 Drawings

Drawing No.	Sheet	Rev.	Date	Title
Not used				

2.3 Attachments

Attachment No.	Rev	Title
Attachment A		Contractors Daily Report
Attachment B		Administrative Procedures
Attachment C		Site Rules (to be provided once agreed with Owner)
Attachment D		Project Health, Safety, and Environmental Rules & Regulations (to be provided once agreed with Owner)
Attachment E		Quality Assurance and Control Procedures (to be provided once agreed with Owner)
Attachment F		Construction / Mechanical Completion (to be provided once agreed with Owner)
Attachment G		Scheduling and Reporting Procedures (to be provided once agreed with Owner)
Attachment H		Specified Requirements

Attachment I		Form of Parent Company Guarantee
Attachment J		Form of Letter of Credit
Attachment K		Projects On Line
Attachment L		SAP Enrolment
Attachment N	Draft for Industry Review, July 7 2008	RTD EMU–DMU Commuter Rail Passenger Railcar Procurement - Technical Provisions
Attachment O		Training

2.4 Exhibits

Exhibit 1	Company's Project Schedule, also referred to as Baseline Schedule

2.5 Manufacturer's Instructions

Contractor shall comply with all Original Equipment Manufacturer's (OEM) directions, instructions and specifications for materials, supply, storage and installation methods, unless otherwise specified.

Contractor shall notify Company immediately, in writing of any conflict between the Project Specifications and these OEM's directions, instructions and specifications. Company will designate which document is to be followed.

3.0 DESCRIPTION OF WORK – SPECIFIC

The Contractor shall deliver vehicles including on-board systems/subsystems that meet the requirements of this Scope of Work and respective specifications. The Contractor shall conduct system engineering activities in cooperation with all system/subsystem suppliers. This shall ensure proper integration of all equipment in a manner that results in compliance with the performance as specified.

The Company has responsibilities for systems integration to achieve a system of Rolling Stock, traction electrical supply, track, signal/train control, communications, facilities, and operations and maintenance. The Contractor is responsible to support the Company with this obligation for the Rolling Stock and the interfaces with other system components, to ensure that all components of the system will function properly and achieve the required performance and safety levels. The Contractor shall manage the Rolling Stock, build, delivery, test and commissioning project, including design process, and quality management processes, ensuring that appropriate engineering processes are employed to develop the products and subsystems/components, validate their individual performance criteria and lead a collaborative process to ensure integration into the vehicle and demonstration of the required performance. This shall include all the Contract Data Requirements List (CDRL Attachment 6 to the Concession Agreement) relating to the Vehicle.

The Contractor shall deliver, to the Company, Rolling Stock, spare parts, special tools, and test equipment, documents and services which meet the physical, performance and documentation requirements in accordance with in Attachment 7 to the Concession Agreement Part C, Rolling Stock Requirements.

The Contractor shall:

- Comply with all Applicable Requirements and with the standards and recommended practices of the following authorities, agencies and organizations:

(i) Federal Railroad Association (FRA) and Federal Transit Administration (FTA);

(ii) the United States Department of Human Services (USDHS);

(iii) American Public Transit Association (APTA);

(iv) Association of American Railroads (AAR);

(v) the State of Colorado;

(vi) the Environmental Protection Agency (EPA);

(vii) National Fire Protection Association (NFPA); and

(viii) Occupational Safety and Health Administration (OSHA)

- Provide technical data and information as needed to support all regulatory processes required to achieve compliance and revenue service at or before Revenue Service Target Date. This applies to the Rolling Stock and Rolling Stock interfaces.
- Provide all the Rolling Stock technical studies, reports, deliverables and simulations requested by the Company
- Provide properly formatted, original input data to the Maintenance Information System data base and continue providing timely updates throughout the warranty period.
- Procure the Automatic Train Control (ATC) and Positive Train Control (PTC) on-board systems from Company's Nominated Supplier, as specified and defined by the Company.
- Integrate the ATC and PTC on-board systems in accordance with Sections 3.18.2 (PTC) and 3.18.3 of this Part 1 Scope of Work.
- Test with OEM test equipment and appropriate OEM technical support for ATC and PTC on-board systems in the Contractor's factory.
- Procure, integrate, and test the Event Recorder and Alerter

Interfaces required with the Owner are to be facilitated and led by the Company. Support may be required from the Contractor in meetings or presentations, with further assistance in preparation of materials, and in development of responses to reasonable requests from the Owner. The Contractor is allowed direct contact with the Owner only with express permission from the Company.

3.1 General Requirements for the Rolling Stock

(a) Rolling Stock shall be capable of reliably providing revenue service in accordance with the operational requirements defined in Attachment 10 to the Concession Agreement (O&M Specifications), including, but not limited to, run times, passenger carrying capacity and the loading standards.

(b) The Rolling Stock will be delivered in accordance with the Company's schedule, Exhibit 1

(c) The Rolling Stock shall comply with Attachment 7 to the Concession Agreement, Part C, Section 1.1 (b) through (f).

3.2 Responsibilities of the Contractor

3.2.1 Rolling Stock Design

(a) The design shall be based on the Silverliner V Cars being designed and produced by the Contractor. Any and all design modifications identified through such activities as initial testing and in-service operation, in particular those relating to safety, performance, operability, and maintainability, shall be presented to the Company. These design modifications shall be incorporated into the base Car design unless otherwise directed by the Company.

(b) The Contractor shall ensure the proper interrelation, functioning and system integration of all Rolling Stock systems and their interrelationships with other parts of the Car and the Commuter Rail Network. The Contractor will support the design development of the maintenance shops, equipment, and infrastructure with respect to providing information pertaining to the Rolling Stock interfaces as detailed in Section 3.2.2.

(c) The Contractor shall submit to the Company design documentation necessary to demonstrate to the Company that the Rolling Stock meets the requirements of the Contract. The Contractor shall demonstrate through the relevant Design Submittals (or through other means acceptable to the Company) for each unique Car type that its Rolling Stock design accounts for each performance, safety, comfort, operation and testing requirement demonstrated in Attachment 7 to the Concession Agreement, Part C and D . This documentation will be used by the Company

to satisfy the requirements of the Owner. The Contractor will support all reasonable requests made by the Company in achieving the appropriate acceptance or approval from the Owner.

(d) The Contractor shall verify the design life for major components and sub-systems as part of the Maintenance Plan. The maintenance documentation provided by the Contractor including Maintenance Allocation Chart, Maintenance Manual and Overhaul Manual shall reflect the expected life of these systems.

3.2.2 INTERFACE MANAGEMENT

(a) The Contractor shall coordinate the design and manufacture of Rolling Stock so as to ensure the Rolling Stock is compatible with the Commuter Rail Network in all respects necessary for operation and maintenance of the Rolling Stock and the Commuter Rail Network.

In addition, the Contractor shall provide timely input into the design process of various interface areas of the Commuter Rail Network relating to vehicle. These areas include:

- the Commuter Rail Maintenance Facility (CRMF) layout, facilities and equipment;
- alignment geometry and clearances;
- trackwork layout and rail profile;
- station platform dimensions and platform edge interfaces;
- TES parameters;
- train control track occupancy shunt resistance parameters and Carborne equipment;
- communications systems, and
- signal systems, including ATC and PTC.

(b) The Contractor shall ensure that Rolling Stock complies with the requirements of Section 3.7 herein, as well as meeting project EMC criteria, as defined in Section 6 (*Electromagnetic Compatibility*) of Attachment 7 to the Concession Agreement, Part A.

(c) The Contractor shall ensure Rolling Stock design is coordinated with the Final Steep Grade Hazard Analysis required in Section 1.2 (*Alignment*) of Attachment 7 to the Concession Agreement, Part B. Any proposed variances in Rolling Stock design and performance from design and performance assumptions stated in the Steep Grade PHA1 to which FRA has concurred shall be submitted to Company for approval by the Owner and FRA for concurrence in accordance with Section 1.2.1 (*Steep Grade Hazard Analysis*) of Attachment 7 to the Concession Agreement, Part B. Rolling Stock performance on steep grades shall be demonstrated through testing.

(d) Steep Grade Preliminary Hazard Analysis is to be updated by the Company. The Contractor shall provide support, including relevant information on air usage, braking forces, etc.

3.2.3 Mock-ups and Samples

(a) The Contractor shall participate, as required, in and provide samples and mock-ups of the Rolling Stock design options for public meetings with regard to public elements of the Rolling Stock, as further defined in Section 9 (*Public Information Program*) of Attachment 9 to the Concession Agreement.

(b) Contractor shall provide design options be in accordance with Sections (b), (c) and (d) of Section 1.2.3 of Attachment 7 to the Concession Agreement, Part C.

3.2.4 ARTIST'S RENDERINGS AND COLOR SAMPLES

(a) The Contractor shall submit interior and exterior overlays and "photo realistic" renderings showing three different color schemes applied to the interior and exterior of the Cars and in accordance with Section 2.5 (*Visual Identity and Branding*) of Part A, of Attachment 7 to

the Concession Agreement. The Contractor shall provide digital images in color and monochrome, and one color board for each rendering. [CDRL #7C-02]

(b) The Contractor shall submit with each rendering a schedule of decorative materials required to implement the rendering including material specification and a physical sample for each such material. The schedule shall include each type of plastic, upholstery material, fabric, paint, floor covering, ceiling panel, interior liner, gasket, and trim, and shall identify the proposed use and location of each material. [CDRL #7C-03]

3.2.5 SEATING MOCK-UP

The Contractor shall submit sample production prototypes of each seat type proposed for the Rolling Stock in the form of a mock-up conveying all aspects of the seat design. The mock-up shall include a sufficient number of rows of each type of transverse mounted seats, bulkhead seats and longitudinal mounted flip-up seats, to demonstrate the relationship of the seats to the side wall, to the aisle and to each other, and to allow evaluation of Passenger comfort. The mock-up shall include a representative sample of each type of seat and each cloth choice. [CDRL #7C-04]

3.2.6 PILOT CARS

Pilot Cars shall be provided in accordance with Section 1.2.6 of Attachment 7, to the Concession Agreement, Part C.

3.2.7 TESTING

The Contractor shall perform or ensure the performance of all Rolling Stock tests, in accordance with Attachment 7 to the Concession Agreement, Part D (*Verification and Demonstration*). Certificates of inspection and testing, and approvals required by Relevant Authorities shall be secured by the Contractor and copies maintained as part of the safety certification documentation required in Section 8.4 (*Safety and Security Certification*) of Attachment 9 to the Concession Agreement.

It is Company's and Contractor's intent to use test evidence previously generated for the SEPTA project, where applicable and subject to approval by the Owner and Concessionaire.

3.2.8 TECHNICAL SUPPORT

Unless other agreed in writing, the Contractor shall have a minimum of three (3) fully qualified technical personnel on-Site beginning with the delivery of the first production Car through Final Acceptance to handle trouble-shooting and to support acceptance testing and commissioning activities.

During the first year following Revenue Service Commencement Date, at least one (1) fully qualified technical person shall be on-Site to continue support of maintenance activities and any modification programs that may be necessary.

During the second and third years after Revenue Service Commencement Date, a fully qualified technical representative will be made available on-Site within 24 hours in case of emergency or severe service impact, otherwise within 72 hours.

From year four (4) through year six (6) after Revenue Service Commencement Date, a fully qualified technical representative will be made available on-Site within 5 business days of receipt of request for service from the Company.

The Contractor will provide Training in accordance with Attachment O Training and Exhibit 1 Company's Project Schedule

3.3 Rolling Stock Drawings

3.3.1 Drawing Database

The Contractor shall maintain an indexed drawing database. The index, and revisions of which, shall be submitted to Company on no less than a quarterly basis until the final Car enters revenue service. [CDRL #7C-05]

3.3.2 Contractor's Drawings

(a) The Contractor shall, within 90 days after the Phase 1 Effective Date, review complete and comprehensive arrangement drawings of each Car type at a Periodic Design Review Meeting. The arrangement drawings shall be dimensioned and shall include the floor plans, reflected ceiling plans, equipment arrangement, equipment compartments, cab layouts, interior longitudinal sections of both sides of the Cars, exterior side elevations of both sides of the Cars, elevation views of both ends of the Cars, and sufficient transverse half-sections or full-sections through the Car to show all variations in cross-section, such as at windows, doors, and door pockets.

(b) The Contractor shall submit a Final Design drawing set that shall include general arrangements, layouts and interface plans for each major Car element, system and subsystem. [CDRL #7C-06]

3.3.3 FRA Safety Compliance Drawings

The Contractor shall submit to the Company detailed design drawings of Rolling Stock safety appliances required by 49 CFR 38 and 49 CFR 231. The drawings shall reference applicable sections of such regulations, indicate the manner of attachment to the Car and cross-reference actual dimensions to those required in such regulations. [CDRL #7C-07]

3.3.4 As-Built Drawings

(a) The Contractor shall submit to Company for review a list and an indexed full set of as-built drawings for the Rolling Stock [CDRL #7C-08]:

- within 40 days after completion of the Pilot Cars,
- upon completion of the last Car, and
- again after completion of any modifications.

The indexed full set of drawings shall be submitted electronically in a recognized drawing format (for example .dxf or .dwg). The list shall be submitted electronically in .pdf format.

(b) The as-built drawings shall include the items referenced in Section 1.3.4 (b) of Attachment 7 to the Concession Agreement, Part C.

3.3.5 Car History Books

(a) The Contractor shall prepare and submit to the Company an electronic copy of the Car History Book for each Car. These shall be in Adobe .pdf format.

(b) Each Car History Book shall provide the information as detailed in Section 1.3.5 (b) of Attachment 7 to the Concession Agreement, Part C.

3.4 DESIGN CRITERIA

3.4.1 ROLLING STOCK DESIGN PARAMETERS

Rolling Stock shall conform to the requirements identified in Section 2.1 of Attachment 7 to the Concession Agreement, Part C

3.4.2 Minimum Design Life

The component, sub-system, and system design life shall be consistent with industry accepted levels. The design, quality, and durability of the Car's components shall be consistent with a minimum design life of 30 years:

[illegible]	[illegible]
Car body	30 years
Truck	30 years
Truck central bearing material	10 years
Gear unit – gears	2,500,000 miles
Gear unit bearings	1,000,000 miles
Seals - Underfloor mounted equipment and enclosure	10 years
HVAC condenser fan motor bearings	70,000 hours
Fluorescent lamps	20,000 hours
Individual LEDs and LED assemblies	100,000 hours
Main transformer cooling fan motor bearing	8 years
Main Transformer	30 years
Brake friction material	33,000 miles (with full dynamic brake performance)
Brake disc	500,000 miles (with full dynamic brake performance)
Compressor Motor bearings	6 years
Air dryer desiccant	3 years
Compressor – Air filters	30,000 miles or 6 months whichever is longer
Elastomeric parts (non friction wear parts such as air spring element)	10 years
Vulcanized elastomeric / metal parts (Subject to friction wear such as bushings for gear unit and motor, shock absorber bushings and lateral bumper rubber etc.)	6 years
Commutating capacitors	20 years
Jumper cables	100,000 cycle flexing test
Electric coupler (automatic coupler)	10,000 coupling / uncoupling
Door crew switches	100,000 cycles

3.4.3 Car Characteristic Requirements

Rolling Stock shall conform to the requirements identified in Section 2.2 of Attachment 7 to the Concession Agreement, Part C

3.4.4 WEIGHT REQUIREMENTS

(a) The Contractor shall develop and provide for review at the first Periodic Design Review Meeting to address Rolling Stock, a Weight and Balance Management Program, including details of weight difference between ends of the Car and the lateral imbalance, and a test procedure and report for determining the center-of-gravity for each Car type. Such weight and balance program shall be updated no less than quarterly as the design progresses with current weight estimates.

(b) The Contractor shall maintain Car weights within those weights specified herein. For weight calculations, passenger weight shall be assumed to be 165 lbs/person.

(c) Each Car shall be balanced laterally and longitudinally. The calculated lateral imbalance shall not exceed 40,000 inch-pounds. The truck-to-truck weight difference within a Car shall not exceed 2,000 pounds. The load on each axle of a Car with AW0 load in a ready to run condition shall be between 24 and 26 percent of the Car weight.

The Car weight and axle load shall not exceed the amounts shown in table below;

Car Weight and Axle Load Table:

Loading Level	Weight (in pounds)		Maximum Axle Load (in pounds)	
	A Car	B Car	A Car	B Car
AW0	138,531	140,000	36,018	36,400
AW1	153,546	155,015	39,772	40,154
AW2	169,221	170,690	43,691	44,073
AW3	176,975	178,444	45,262	46,011
AW4	184,896	186,365	47,609	47,991

3.5 Dynamic Performance / Operating Requirements

3.5.1 General Performance Requirements

(a) The Contractor shall demonstrate via simulation that the proposed Rolling Stock will meet the minimum service standards required in Attachment 10 to the Concession Agreement, (*O&M Specifications*) taking into account end-of-route turn times, required headways and service levels, and any pertinent track-occupancy issues in its calculations.

(b) The Contractor shall submit the ***Rolling Stock Performance Report*** demonstrating the Rolling Stock's ability to comply with the performance requirements contained in Attachment 10 to the Concession Agreement, (*O&M Specifications*), the run time and energy consumption requirements of Attachment 19 to the Concession Agreement, (*Concessionaire's Proposal*) and the requirements of Attachment 7 to the Concession Agreement, Part C. [CDRL #7C-10]

The Rolling Stock Performance Report shall include the requirements as identified in 2.4.1. (b) (i) through (iii) of Attachment 7 to the Concession Agreement, Part C.

(c) The Contractor shall review preliminary design documentation leading to the completion of the Rolling Stock Performance Report and the Duty Cycle Report detailed in Section 2.4.5 of Attachment 7 to the Concession Agreement, Part C at Periodic Design Review Meetings.

(d) The Contractor shall provide an initial energy consumption model within 30 days following the issuance of the Notice to Proceed, simulating a vehicle loaded to AW2 for one round trip on each line. The Contractor shall develop an energy management plan, incorporating weight management, to monitor the projected energy use of the vehicle. The Contractor shall comply with the initial energy consumption model and verify it through tests to be conducted prior to the Revenue Service Commencement Date on each Commuter Rail line. The Company will provide operating staff for these tests.

3.5.2 Service Braking

The service braking system shall comply with the requirements of Section 2.4.2 of Attachment 7 to the Concession Agreement, Part C.

3.5.3 Emergency Braking

The emergency braking system shall comply with the requirements of Section 2.4.3 of Attachment 7 to the Concession Agreement, Part C.

3.5.4 Dynamic Braking

The Cars shall have an electric dynamic brake system and automated controls to blend dynamic and friction braking. The Contractor shall select the level of dynamic brake performance based on the propulsion system design, but not to exceed the service brake performance requirements set forth in Section 2.4.2 of Attachment 7 to the Concession Agreement, Part C.

3.5.5 Duty Cycle

(a) The Rolling Stock and all its systems shall be capable of continuous normal service operation over the Commuter Rail Network loaded to AW2.

(b) The Contractor shall provide to Company a report (the ***Duty Cycle Report***) that demonstrates that all systems are capable of such continuous normal service operation within their respective rated thermal, electrical and mechanical values, as well as details of the equipment failure scenario duty cycles that can be tolerated, as further defined below. [CDRL #7C-12]

(c) The Duty Cycle Report shall satisfy the requirements of Section 2.4.5. (c) of Attachment 7 to the Concession Agreement, Part C.

(d) The Contractor shall test the Rolling Stock to prove the normal and rescue duty cycle capabilities specified in the Duty Cycle Report.

(e) The friction brakes shall be capable of providing all of the service braking required at any speed or Car load, if the electric brakes fail.

3.5.6 Jerk Limiting

Changes in tractive effort shall be jerk limited. The jerk limit rate shall comply with the requirements of Section 2.4.6 of Attachment 7 to the Concession Agreement, Part C.

3.5.7 Ride Quality

Rolling Stock shall provide ride quality compliant with the requirements of Section 2.4.7 of Attachment 7 to the Concession Agreement, Part C.

3.6 Noise and Vibration Requirements

3.6.1 Noise Criteria

The Car shall comply with the noise criteria requirements of Section 2.5.1 (a) and (b) of Attachment 7 to the Concession Agreement, Part C.

(a) The Contractor shall submit a report (the ***Rolling Stock Noise Calculations Report***) detailing the estimated noise levels developed during the Car design process and demonstrating that the design will meet the requirements of this Section 2.5 of Attachment 7 to the Concession Agreement, Part C. [CDRL #7C-12]

(b) The Contractor shall maintain the Rolling Stock Noise Calculations Report throughout the design process and test the performance estimates therein.

3.6.2 Interior Noise Levels

3.6.2.1 Cab Noise Levels

The Car shall comply with the cab noise requirements of Section 2.5.2.1 Attachment 7 to the Concession Agreement, Part C.

3.6.2.2 Passenger Area Noise Levels

The Car shall comply with the passenger area noise requirements of Section 2.5.2.2 Attachment 7 to the Concession Agreement, Part C.

3.6.3 Wayside Noise

The Car shall comply with the wayside noise requirements of Section 2.5.3. Attachment 7 to the Concession Agreement, Part C.

3.6.4 Vibration Criteria

The Car shall comply with the vibration criteria requirements of Section 2.5.4 Attachment 7 to the Concession Agreement, Part C.

3.7 Electromagnetic Compatibility

3.7.1 General Requirements

(a) Cars shall be electromagnetically compatible within themselves, with other trains operating on the Commuter Rail Network, with signal systems, with communications systems, and other electronic equipment and with equipment owned and operated within EMI range of the Commuter Rail Network. The Cars shall not be capable of any operation that causes unsafe conditions in the signal systems.

(b) The Contractor shall supply documentation relating to Rolling Stock compliance with the requirements of Section 2.6 of Attachment 7 Part C for inclusion within the Company's overall EMC Control Plan and EMC Test Program, as set forth in Section 6 (Electromagnetic Compatibility) of Attachment 7 Part A and in accordance with the APTA PRESS Standard for the Development of an Electromagnetic Compatibility Plan SS-E-010-98.

(c) Rolling Stock shall meet the requirements of this Section 2.6 of Attachment 7 to the Concession Agreement, Part C under any operating condition.

(d) In the event of electromagnetic incompatibility or interference between systems, either within the vehicle, or between the vehicle and non-vehicle systems (for example, wayside equipment, signal system), the Contractor and all applicable subcontractors / suppliers shall work jointly with all parties to resolve the issue in a collaborative manner.

3.7.2 Induced Emissions

The induced emissions shall comply with the requirements of Section 2.6.1.2 of Attachment 7 to the Concession Agreement, Part C.

3.7.3 Conducted Emissions

The conducted emissions shall comply with the requirements of Section 2.6.1.3 of Attachment 7 to the Concession Agreement, Part C.

3.7.4 Radiated Emissions

The radiated emissions shall comply with the requirements of Section 2.6.1.4 of Attachment 7 to the Concession Agreement, Part C.

3.7.5 Cab Signal Interference

The cab signal interferences shall comply with the requirements of Section 2.6.1.5 of Attachment 7 to the Concession Agreement, Part C.

3.7.6 Subsystem EMC Requirements

Subsystem EMC shall comply with the requirements of Section 2.6.2 of Attachment 7 to the Concession Agreement, Part C.

3.8 Equipment Configuration / Redundancy

Equipment configuration shall provide fault tolerance and back-up capabilities as required in Section 2.7 of Attachment 7 to the Concession Agreement, Part C. ATC (Alternate Technical Concept) submission DTP-ATC-011, as approved by Owner, applies to this section.

3.8 CARBODY STRUCTURE

3.8.1 General carbody Requirements

The carbody structure shall comply with the requirements of Section 3.1 of Attachment 7 to the Concession Agreement, Part C except for the use of sealants which may be used in the interior of the Car, as described in the DTP-ATC-0027.

3.8.2 Carbody Strength

The strength of the carbody shall comply with the requirements of Section 3.2 of Attachment 7 to the Concession Agreement, Part C.

3.8.3 Stress Analysis

The Contractor shall certify that the carbody structure and all appurtenances comply with the requirements of APTA PRESS Standard for the Design and Construction of Passenger Railroad Rolling Stock SS-C&S-034-99.

3.8.4 Carbody Structural Submittals

3.8.4.1 Structural Design Load Report

(a) The Contractor shall submit a report (the ***Carbody Compliance Report***) demonstrating compliance with the requirements of 49 CFR 238 and APTA PRESS Standard SS-C&S-034-99, Rev. 2, and detailing the characteristic deformation of the carbody structural assembly. [CDRL #7C-13]

(b) The carbody Compliance Report shall fulfill the requirements of Section 3.4.1. (b) of Attachment 7 to the Concession Agreement, Part C.

3.8.4.2 Collision Energy Management and Survivability Analysis Report

The Contractor shall submit a report (the ***Collision Energy Management and Survivability Analysis Report***) consistent with the requirements of APTA PRESS Standard SS-C&S-034-99, Rev. 2, identifying the contributions made by the coupler and each structural member, to collision energy absorption within the carbody. The Collision Energy Management and Survivability Analysis Report shall specify the level of collision energy management necessary to ensure compliance with the safety of Passengers during a collision as required by the Hazard Identification Analysis and Resolution Process as set forth in Section 8 (*Hazard and Threat Management*) of Attachment 9 to the Concession Agreement, . [CDRL #7C-14]

3.9 CARBODY FEATURES

3.9.1 Industrial Design

(a) The Contractor shall develop the overall concept design of the Car and its arrangement so as to optimize the function and appearance of the Car for the mutual benefit of Owner and passengers.

3.9.2 Train Egress Calculation

(a) The Contractor shall submit a calculation (the Train Egress Calculation) that shall provide the time to evacuate passenger loads for AW1, AW2 and AW3 (including wheelchair Passengers), based on the arrangement of seats, doors, and any interior steps on the Rolling Stock; and shall provide justification that this duration is reasonable and in accordance with Good Industry Practice for service operations and emergency scenarios. The calculation shall consider both egresses at stations and on the wayside. [CDRL #7C-15]

(b) The Train Egress Calculation shall be used for determining dwell times at intermediate stations, as may be included in planning the Operating Timetable.

3.9.3 Accessibility Provisions

The Rolling Stock accessibility provisions shall comply with the requirements of Section 4.3. of Attachment 7 to the Concession Agreement, Part C.

3.9.4 Interior Construction

The interior construction shall comply with the requirements of Section 4.4. Of Attachment 7 Part C. ATC (Alternate Technical Concept) submission DTP-ATC-024, applies to section 4.4.1 of Attachment 7 to the Concession Agreement, Part C.

3.9.5 Passenger Seats

The passenger seats shall comply with the requirements of Section 4.5 of Attachment 7 to the Concession Agreement, Part C. ATC (Alternate Technical Concept) submission DTP-ATC-023, applies to this section.

3.9.6 Doors

The Contractor shall provide design documentation for review at Periodic Design Review Meetings that defines details of each door and door system on the Cars and demonstrates how the designs meet the requirements of Section 4.6 of Attachment 7 to the Concession Agreement, Part

C and provides functionality and reliability for the Rolling Stock in accordance with such requirements of Section 4 of Attachment 7 to the Concession Agreement, Part C.

3.9.6.1 Side Doors

The side doors shall comply with the requirements of Section 4.6.1 of Attachment 7 to the Concession Agreement, Part C.

3.9.6.2 Collision Post Doors

The collision post doors shall comply with the requirements of Section 4.6.2 of Attachment 7 to the Concession Agreement, Part C.

3.9.6.3 Automatic Passenger Counters

(a) Rolling Stock shall be fitted with an automatic passenger counting (APC) system in accordance with requirements of Section 4.6.3 of Attachment 7 to the Concession Agreement, Part C.

(b) The Contractor shall assist in implementing an INIT APC garage server at the CRMF for downloading data collected by the APCs on the Rolling Stock using IEEE 802.11n standard and Wi-Fi Protected Access (WPA2) security standard with AES encryption and 802.11i security standards. Data from the garage servers shall then automatically transfer this data to RTD in the INIT APC standard format using secure FTP server (SFTP), not FTPS, through the Company's own secure broadband internet connection to Owner's SFTP server at least once per day.

3.9.7 Windows

All exterior windows of the Car shall meet the requirements of 49 CFR 238.221 and 223.

3.9.8 Car End Equipment

3.9.8.1 Buffer Assembly

The buffer assembly shall comply with the requirements of Section 4.8.1 of Attachment 7 to the Concession Agreement, Part C.

3.9.8.2 Diaphragms

The Car end diaphragms shall comply with the requirements of Section 4.8.2 (a) and (b) of Attachment 7 to the Concession Agreement, Part C.

The Contractor shall review at Periodic Design Review Meetings design documentation providing details of diaphragm design, size, arrangement, and installation, as well as any necessary exterior coating.

3.9.8.3 Inter-Car Barriers

The inter-Car barriers shall comply with the requirements of Section 4.8.3. (a) through (d) of Attachment 7 to the Concession Agreement, Part C.

The Contractor shall provide a review at Periodic Design Review Meetings of details of the design, arrangement, and installation of the inter-car barriers.

3.9.10 Over-Window Luggage Racks

The Contractor shall provide over-window luggage racks in high ceiling areas, which shall include longitudinal and lateral restraints for stowed articles. The racks, including attachments to the carbody, shall have an ultimate strength sufficient to comply with the requirements of 49 CFR 238.233 and APTA Standard SS-C&S-006-98.

3.9.11 Lettering, Signs and Numbers

3.9.11.1 General Identification Requirements

(a) The Contractor shall provide decals and signs on Rolling Stock in accordance with applicable Law and Section 4.10 of Attachment 7 to the Concession Agreement, Part C. Signage, decals, graphics, plates, numbers, and instructions for egress/access of Passengers shall each comply with 49 CFR 27, 37, and 38, APTA PRESS Standards SS-PS-002-98 and SS-PS-004-99.

(b) The Contractor shall provide an active Low Location Exit Path Marking system as defined in APTA PRESS Standard SS-PS-004-99.

3.9.11.2 Destination Signs

The Contractor shall provide Cars with destination signs in compliance with the requirements of Section 4.10.2. (a) through (c) of Attachment 7 to the Concession Agreement, Part C.

3.9.11.3Route Map Frames

The Contractor shall provide a rectangular, one-piece frame with no visible welds or seams sized to accept standard 21 x 33-inch route maps on the bulkheads of each Car. The color and finish of such frames shall be in accordance with the color scheme selected by the Company in accordance with Section 1.2.3. (b) of Attachment 7 to the Concession Agreement, Part C.

3.9.12 Exterior Styling

3.9.12.1 Exterior Graphic Package

The Contractor shall provide and apply an exterior graphics package to the carbody. The exterior graphics package shall include all FRA and APTA required and recommended signage, the Owner logo, and any decals within the approved exterior color scheme.

3.9.12.2 Car Number Sign

The Contractor shall provide exterior, permanent metal Car number signs on the cab-end of each cab Car.

3.9.13 Emergency Equipment

Each Car shall have emergency equipment, including:

(a) auxiliary portable lighting that complies with 49 CFR 239;

(b) a pry bar complying with the requirements of 49 CFR 239; and

(c) one 20-pound, A/B/C rated, dry chemical fire extinguisher, marked with "PROPERTY OF RTD" in 1-inch high letters on the extinguisher case.

(d) an emergency egress ladder / walkway as detailed below, unless an alternate method for emergency egress to ground level is provided that permits vehicle evacuation in appropriate time frames:

a. A combination emergency ladder/walkway device shall be provided in each Car to facilitate passenger de-training to ground level and also onto an adjacent rescue train (alongside or at either end).

b. The ladder/walkway shall be a two-piece assembly made of high-strength, nonconductive material with each piece weighing no more than 40 pounds and the total assembly weighing no more than 50 pounds. One piece of the assembly shall serve as the ladder and the other shall readily attach (without the use of tools) and be secured to the ladder to create a walkway. The device shall safely handle a 450-pound person when used as a ladder and as a walkway. When used as a walkway, the device shall span seven feet.

c. The ladder/walkway shall be stored and easily removable from the emergency equipment locker located inside the Car near or adjacent to the F end vestibule side body doors.

d. The design of the ladder/walkway shall permit passenger egress through the body end doors as well as through the side body entrance doors down to track level, or onto a rescue train. When used as a walkway, provisions for securing to both Cars shall be provided.

e. The ladder shall be of a compact design that is easy to deploy and made for safe, effective and intuitively obvious use by passengers.

f. For end door use, the ladder shall rest on the track bed which is approximately 7 inches below TOR. The ladder shall be designed to rest securely on wood, concrete and track bed ballast without requiring any additional effort or adjustment.

g. For side door use, the ladder shall be fully supported by, and secured to, the carbody and not require contact with the ground for usage. For side door use, the ground height will vary by location, and will vary between TOR and 24 inches below TOR. The ladder shall have provisions for adjusting to differing elevations of the Car with respect to the ground.

h. The treads of the ladder shall be at least 16 inches wide, with a step tread depth of at least 3 inches, and shall incorporate a permanent non-skid surface or coating. In

addition, when deployed at both the body end doors and the side doors, there shall be at least 8 inches of toe clearance on all treads.

i. The surface of the walkway shall be at least 24 inches wide and shall incorporate a permanent non-skid surface or coating on the side exposed to foot traffic.

j. The ladder shall be permanently marked with 2-inch high letters "PROPERTY OF RTD".

3.9.14 Safety Appliances

Safety appliances shall meet the requirements of Section 4.13 of Attachment 7 to the Concession Agreement, Part C.

3.9.15 Horn

The Contractor shall provide each cab-Car with a roof-mounted horn at the cab end, protected from ingress of snow and rain. The horn sound level shall comply with applicable Law including the requirements of 49 CFR 229.129.

3.9.16 Bicycle and Oversize Luggage Storage

Bicycle and oversize luggage storage shall meet the requirements of Section 4.15 of Attachment 7 to the Concession Agreement, Part C

3.9.17 Fuse/Torpedo Container

The Contractor shall provide a container for carrying fuses and torpedoes compliant with 49 CFR 229.119(f).

3.10 OPERATOR'S CAB

3.10.1 General Cab Requirements

(a) The Contractor shall provide an operator's cab at the front of each Car, with the operator located on the right side of the Car. Each cab shall have a locking door.

(b) Each operator's cab shall be identical with the exception of related controls specific to the Car type. Each cab shall be equipped with an operator's console and all equipment necessary for the operation of the train. The cab and operator's console shall be ergonomically compatible with adults from the 5th-percentile female to the 95th-percentile male.

(c) The Contractor shall comply with the requirements of applicable Law, including 49 CFR 229 and 238 and the guideline of DOT/FRA/ORD-98/03 Human Factors Guidelines for Locomotive Cabs.

(d) The Contractor shall ensure that the operating philosophy of Company is addressed through the cab layout, including:

(a) Location of door controls for both sides of the cab,

(b) Sliding exterior windows on both sides of the cab that enable the operator to look out and check door closing,

(c) Locating PTC and CCTV screens within the cab console, and removing superfluous equipment relating to the existing design,

(e) The Operator's desk shall include an integrated power and brake master controller of proven design to replace the separate power and brake controllers in the Silverliner V design.

(f) Installation of a single person flip up seat in the non-operator side of the cab for use by training or supervisory staff.

(g) In multiple EMU operation, the cab partition door in the cab shall permit the lock out of the driver side of the cab to create a walk through the cab to permit movement between married pairs through the collision post doors of each coupled Car, but not allowing passengers to enter or occupy of the cab driver area. Non cab side controls should be non-functional, hidden, with a locked cover, from view when not in operational use. The non driver window should also be lockable.

3.10.2 Cab Controls

Cab controls shall comply with Section 5.2 of Attachment 7 to the Concession Agreement, Part C.

A separated heater shall be installed in the cab with local control.

3.10.3 Cab Design Collaboration

The Contractor shall establish a process of collaboration in the detailed design of the cab to meet the needs of the Company and the requirements set forth in the Agreement and ensure that representatives of the O&M Contractor participate. No later than the conclusion of the Preliminary Design of the cab, the Company shall invite Owner to attend and participate in meetings where cab design documentation is under collaborative review.

The cab layout shall support the operating philosophy of the Company and shall be agreed with the Company's operator representatives meeting their ergonomic and operational requirements.

3.11 HEATING, VENTILATION AND AIR CONDITIONING

3.11.1 System Requirements

The Rolling Stock shall have an integrated heating, ventilation and air conditioning system (HVAC) compliant with the requirements of Section 6.1 of Attachment 7 to the Concession Agreement, **Part** C.

3.11.2 Required Interior Conditions

The HVAC system shall maintain interior temperatures in accordance with the requirements of Sections 6.2.1 and 6.2.2 of Attachment 7 to the Concession Agreement, Part C.

3.11.3 HVAC Performance Report

The Contractor shall submit a report (the ***HVAC Performance Report***) demonstrating compliance with the requirements of Section 6. of Attachment 7 to the Concession Agreement, Part C. [CDRL #7C-16]

The HVAC Performance Report shall include the content identified in Section 6.3 (a) through (e) of Attachment 7 to the Concession Agreement, Part C.

3.12 LIGHTING SYSTEMS

(a) The Contractor shall provide a lighting scheme with an emphasis on direct lighting of the passenger seating areas with reference to RTD approval of DTP-ATC-011.

(b) The Car lighting shall comply with the requirements of Section 7 (b) and (c) of Attachment 7 to the Concession Agreement, Part C.

3.13 PROPULSION AND DYNAMIC BRAKING

3.13.1 Propulsion and Dynamic Braking System Documentation

The Contractor shall provide documentation for review at Periodic Design Review Meetings describing the propulsion and dynamic braking system. Such documentation shall include the requirements identified in Section 8.1 (a) through (e) of Attachment 7 to the Concession Agreement, Part C.

3.13.2 Traction Control

Traction control shall comply with the requirements of Section 8.2 of Attachment 7 to the Concession Agreement, Part C. In addition, a Car wash function shall be provided controlling the train consist at a preset speed. The speed shall be variable between 0.5 and 5 mph, and shall be set at 0.5 mph.

3.13.3 Direction Control

Direction control shall comply with the requirements of Section 8.3 of Attachment 7 to the Concession Agreement, Part C.

3.13.4 Load Weigh Compensation

Load weigh compensation for traction effort shall comply with the requirements of Section 8.4 of Attachment 7 to the Concession Agreement, Part C.

3.13.5 Wheel Slip / Slide

Wheel slip / slide shall comply with the requirements of Section 8.5 of Attachment 7 to the Concession Agreement, Part C.

3.13.6 Odometer

Each Car shall be equipped with an odometer to measure Car mileage. It shall not be possible to reset the odometer except by special means.

3.13.7 Pantograph

Pantographs shall be compliant with the requirements of Section 8.7 of Attachment 7 to the Concession Agreement, Part C.

The pantograph contact strip shall be 100% carbon.

3.13.8 Lightning Protection

Cars shall be compliant with the requirements of Section 8.8 of Attachment 7 to the Concession Agreement, Part C.

3.13.9 Safety Grounding

All equipment on the Car shall be safety-grounded to the Car structure.

The Contractor shall provide a grounding plan and scheme that shall become part of the design reviews and required safety report.

3.13.10 Traction Electrification System (TES)

The Rolling Stock shall operate using power from the TES via the pantograph, with return current through the running rails. The TES supply shall be at a nominal voltage of 27 kV AC, with a range between 20 KV AC and 29 KVAC. The supply frequency shall be single phase, 60 Hz.

The Contractor shall ensure that the Car provides a traction power return path affording maximum safety to operating personnel and Rolling Stock equipment protection including axle brushes to protect the wheel bearings

3.13.11 Regenerated Power

The Car shall be capable of providing regenerative braking up to a maximum voltage of 30KV. Regenerated power created under dynamic braking is to be used in the following priority:

(i) By the Car, in systems such as auxiliary power, and HVAC.

(ii) Fed back into the overhead contact wire for use, depending on receptivity.

(iii) Discharged through braking resistors on the Car

3.14 AUXILIARY POWER SYSTEM

3.14.1 Auxiliary Power System Requirements

The auxiliary power system shall comply with the requirements of Section 9.1 of Attachment 7 to the Concession Agreement, Part C.

The Contractor shall ensure that the Auxiliary Power supply is sized to include at least 10% spare capacity above that needed to operate the Cars within the specified Duty Cycle.

3.14.2 Auxiliary Power System Documentation

The Contractor shall review at Periodic Design Review Meetings documentation of the auxiliary power system. Such documentation shall include the requirements as identified in Section 9.2 (a) and (b) of Attachment 7 to the Concession Agreement, Part C.

3.15 TRUCKS AND SUSPENSION

3.15.1 Truck and Suspension Requirements

Trucks and suspension shall meet the requirements as identified in Section 10.1 of Attachment 7 to the Concession Agreement, Part C.

3.15.2 Truck Assembly Finite Element Analysis Report

The Contractor shall prepare a report (the ***Truck Assembly Finite Element Analysis Report***) that shall demonstrate the capabilities of the truck with regard to static and fatigue strength of the truck frame, bolster (if used), equipment mounting brackets, and axles; and shall detail the predicted and designed static and dynamic loads, and the margins of safety for the truck frame, bolster (if used) and axle.

3.15.3 Suspension Behavior Calculation

The Contractor shall prepare a calculation (the ***Suspension Behavior Calculation***) that shall detail the findings of an analysis, using a validated mathematical model, of the quasi-static and dynamic behavior of the Rolling Stock to demonstrate good ride characteristics and stability, curving and safety against derailment.

3.16 AIR SUPPLY AND FRICTION BRAKE SYSTEMS

3.16.1 General

(a) The air supply and brake system shall comply with Section 11.1 (a) through (e) of Attachment 7 to the Concession Agreement, Part C.

(b) The Contractor shall include a failure mode, effects, and criticality analysis (FMECA) of the friction brake system in failure mode in the Rolling Stock Performance Report as set forth in Section 2.1 (*Rolling Stock Design Parameters*) of Attachment 7 to the Concession Agreement, Part C.

3.16.2 Air Supply and Friction Brake System Documentation

The Contractor shall prepare documentation of the air supply and friction brake system for review at Periodic Design Review Meetings that includes the requirements of Section 11.2 (i) through (iii).

3.16.3 Braking Functionality

The braking functionality of the Rolling Stock shall comply with the requirements of Section 11.3 of Attachment 7 to the Concession Agreement, Part C except Emergency braking (Emergency braking shall be controlled at a minimum on a per-Car basis and the emergency brake cylinder pressure shall be limited to AW3 value,)

3.17 Carborne Communications Systems

3.17.1 Description

The Contractor shall provide a complete and functional communication system for each Car, comprised of the components detailed in Section 12.1 (a) of Part C of Attachment 7 to the Concession Agreement

(b) The communications equipment shall not pose any health hazard to the public, Passengers, or operators and shall be in strict compliance with applicable Law, including FRA, AAR and FCC regulations, and state and Federal guidelines, including 49 CFR Part 220

(*Railroad Communications*). Exposure to radio frequency emissions shall conform to IEEE C95.1, Table 2.

(c) The Contractor shall provide documentation of the communications system and its major components for review at Periodic Design Review Meetings that includes:

- a narrative description of the overall system architecture; and
- a comprehensive analysis of how such system design meets the requirements of Attachment 7 to the Concession Agreement, Part C.

3.17.2 Video Surveillance and Recording System

(a) The Contractor shall provide on each Car or minimum operable train unit a video surveillance and recording system, including cameras, a digital video recorder (DVR), CCTV controller and other necessary ancillary equipment.

(b) The requirements of Section 12.2 (b) through (e) of Attachment 7 to the Concession Agreement, Part C shall be met.

(c) An additional camera shall be provided in each cab providing coverage of the operator desk area.

(d) All images should be viewable, realtime, on a monitor screen located in the cab.

(e) Two external cameras, on either side of the Car, viewing the doors and car/platform interface shall be included in the video surveillance and recording system. The external camera images shall be automatically displayed on the driver's monitor on the door open command and remain until the doors are closed and the train is clear of the station platform.

(f) Contractor shall include aerial and transmission equipment to allow recorded and live images to be transmitted to wayside transmission equipment, along the Commuter Rail Network and in the CRMF, as specified by the Company. This can be achieved by integration with the PTC/ATC system.

3.18 OPERATIONAL SAFETY SYSTEMS

3.18.1 General

The Rolling Stock and its safety systems shall comply with the requirements of Section 13 of Attachment 7 to the Concession Agreement, Part C.

The Contractor shall prepare documentation of each operational safety system defined in Section 13 of Attachment 7 Part C for review at Periodic Design Review Meetings, in sufficient detail to demonstrate compliance with each of the requirements set forth in paragraph (a) of Section 13.1 of Attachment 7 to the Concession Agreement, Part C.

3.18.2 Positive Train Control System

The Contractor will supply, integrate, fit, and test the PTC system as specified and defined by the Company, and supplied by a third party (Nominated Supplier) to each minimum operable train consist. The PTC system is to be independent from the Automatic Train Control (ATC) system as set forth in Section 3.18.3 below.

The PTC system shall include a control unit, an onboard display unit in each cab, and propulsion and brake system interfaces and shall be procured as follows:

On Board Positive Train Control (PTC) System – Major Components

Major Components	Supplier
On-board Equipment Rack	Contractor
Equipment Rack Interface Connections	Contractor
PTC Display Unit	Nominated Supplier
PTC Display Unit Connections and Mounting	Contractor
Acknowledge Switches / MMI interface	Nominated Supplier
PTC Train Management Computer	Nominated Supplier
PTC Speed Sensors with cables and Connectors	Nominated Supplier
Speed Sensor Mounting	Contractor
Antennas / Transceivers	Nominated Supplier
Antenna / Transceiver connections and Mounting	Contractor
Power Supply for battery voltage (37.5V)	Contractor

The display unit and MMI shall be located in the cab as agreed by the Company. Interfaces with the propulsion and brake system shall, at minimum, allow the cab signal system as set forth in Section 9.8 (*Cab Signals*) of Attachment 7 to the Concession Agreement, Part B to cut out tractive effort and apply a penalty brake.

3.18.3 Automatic Train Control System

The ATC system will comply with the requirements of Section 13.3 of Attachment 7 to the Concession Agreement, Part C. The Contractor will supply, integrate, fit, and test the ATC system as specified and defined by the Company, and supplied by a third party (Nominated Supplier).

The ATC system shall include a control unit, an onboard cab aspect display unit in each cab, and propulsion and brake system interfaces and shall be procured as follows:

Automatic Train Control (ATC) System – Major Components

Major Components	Supplier
On-board Equipment Rack	Contractor
Equipment Rack Interface Connections	Contractor
Aspect Display Units	Nominated Supplier
Aspect Display Unit Connections and Mounting	Contractor
Acknowledge Switches / MMI interface	Nominated Supplier
Speed Sensors with cables and connectors, the cable length will be defined at detail design stage	Nominated Supplier
ATC Speed Sensor Mounting	Contractor
ATC Train Management Computer	Nominated Supplier
Track Receiver Coils	Nominated Supplier
Track Receiver Coils Connections and Mounting	Contractor
Track Receiver Coil Junction Boxes	Contractor
Power Supply for battery voltage (37.5V)	Contractor

Interfaces with the propulsion and brake system shall, at minimum, allow the cab signal system as set forth in Section 9.8 (*Cab Signals*) of Attachment 7 to the Concession Agreement, Part B to cut out tractive effort and apply a penalty brake.

3.18.4 Alerter System

The Contractor will provide an Alerter System compliant with the requirements of Section 13.4 of Attachment 7 to the Concession Agreement, Part C.

3.18.5 Event Recorder System

(a) The event recorder system (***ERS***) shall comply with the requirements of Section 13.5 (a) and (c) through (e) of Attachment 7 to the Concession Agreement, Part C.

(b) One ERS shall be provided on each Car. The Contractor may provide one ERS per minimum operable train consist if minimum functionality and signal integrity are maintained. The ERS shall be a stand-alone device, independently operated, and powered from the Car's low voltage power supply. The ERS shall be a solid-state microprocessor-based unit. The data storage memory module shall be FRA- and IEEE-compliant. The system shall include any required sensors, transducers, pressure switches, power supplies, and isolation module.

(c) Each Car shall be equipped with an odometer to measure Car mileage as part of the Event Recorder System. It shall not be possible to reset the odometer except by special means.

3.18.6 Speed Sensing System

A speed sensing system shall be provided compliant with the requirements of Section 13.6 of Attachment 7 to the Concession Agreement, Part C.

3.18.7 Train-to-Wayside Communication (TWC) and Wayside-to-Train Communication (WTC)

The Contractor shall be provided with TWC and WTC system that is supplied by Nominated Supplier to transmit the data.

3.18.8 Radio

The Contractor shall be provided with the Radio that is supplied by Nominated Supplier.

3.19 TRAINLINES AND DATA NETWORKS

3.19.1 General

(a) Cars shall comply with the requirements of Section 14.1 (a) through (c) of Attachment 7 to the Concession Agreement, Part C.

(b) The Contractor shall prepare and review at Periodic Design Review Meetings documentation of the trainline approach, defining which functions are implemented using discrete trainlines and which use the data networks. Documentation shall include:

a description of the data networks architecture, control response times and failure scenarios; and

equipment details and the coupling arrangement between Cars.

(e) The Contractor shall demonstrate previous service experience for each control signal proposed to be networked, to confirm the safety and reliability of this approach.

3.19.2 System Architecture

3.19.2.1 General

The System Architecture shall comply with Section 14.2 of Attachment 7 to the Concession Agreement, Part C.

3.19.3 Global Positioning System Receiver

The Cars shall be fitted with Global Positioning System (GPS) receivers for PTC and/or train to wayside communications, in conjunction with section 3.18.2 and 3.18.3, there shall be an antenna on each Car. Such GPS receivers shall be equipped with a Wide Area Augmentation System (WAAS) and differential beacon capability. The system shall utilize the GPS mode with the lowest dilution of precision available at each moment in time.

3.20 MONITORING AND DIAGNOSTICS

(a) Each Car shall include an integrated Monitoring and Diagnostic System (an MDS). The MDS shall be a physically-distributed, functionally-integrated system which monitors signals and events within the Car and within selected subsystems on the Car and stores the collected subsystem, Car, and train data in non-volatile memory within the selected subsystems, at a central data storage point on the Car, and in computer systems located at the CRMF.

(b) The Contractor shall provide each Car with a Local Diagnostic and Test System (LDTS) for each intelligent subsystem and with a central diagnostic system, which shall integrate the individual LDTS of the various subsystems and sensors on the Car.

(c) The Contractor shall review at Periodic Design Review Meetings documentation of the MDS. Such documentation shall include a narrative description and technical details of the overall system architecture and major components, including functionality locally and in the operator's cab.

(d) The MDS shall record data identifying the approximate location of the train on the system whenever failure events occurred.

3.21 MATERIALS AND WORKMANSHIP

3.21.1 General

(a) Materials and workmanship shall be in accordance with the requirements of applicable Law and Good Industry Practice.

(b) The Contractor shall ensure that Materials Safety Data Sheets (MSDS) are prepared for materials used in the fabrication of the Cars, including lubricants but excluding non-hazardous metallic materials. Information in each MSDS shall be in a form compliant with ANSI Z400.1-2004.

3.21.1.1 Quality

The Contractor shall ensure through the Quality Management System set forth in Section 7 (*Quality Management*) of Attachment 9 to the Concession Agreement, that all aspects of the Rolling Stock are in conformance with the design, materials, and workmanship requirements defined in Attachment 7 to the Concession Agreement, .

3.21.1.2 Prohibited Materials

Prohibited materials are detailed in Section 16.1.3 of Attachment 7 to the Concession Agreement, Part C.

3.21.2 Flammability, Smoke Emission and Toxicity Requirements

3.21.2.1 General

(a) All combustible materials used in the construction of the Cars shall satisfy the flammability, toxicity and smoke emissions requirements of 49 CFR 238.103, and this Section 16.2 of Attachment 7 to the Concession Agreement, Part C, and the standards of NFPA 130. In the case of any conflict between such requirements, the most restrictive requirement shall prevail.

(b) The Contractor shall comply with all provisions of 49 CFR 238.103 (c), Fire Safety Analysis for Procuring New Passenger Equipment, and APTA RP-PS-005-00, Fire Safety

Analysis of Existing Passenger Rail Equipment. The Contractor shall submit a Fire Safety Analysis (the ***Fire Safety Analysis***) of the design and materials of construction of the Car, as described in Section 16.2.6 of Attachment 7 to the Concession Agreement, Part C.

(c) The Contractor shall submit to Company test reports from an independent laboratory indicating successful testing and demonstrating compliance with such requirements. Testing shall be conducted after the Phase 1 Effective Date and on a production batch of material intended to be used on the Car. Each laboratory shall have tested a standard test sample no greater than 30 days prior to performing the tests, the results of which shall be submitted to Company.

3.21.2.2 Flammability and Smoke Emission.

The Contractor shall submit to Company for approval a list of materials intended to be used in the Car that are not, in the Contractor's opinion, combustible. [CDRL #7C-18] The Contractor shall test combustible materials used in Cars to demonstrate compliance with the requirements set forth in 49 CFR 238.103 and NFPA 130.

3.21.2.3 Floor Assembly Fire Resistance Testing Criteria

The Contractor shall test the floor assembly in accordance with ASTME 119 to demonstrate a thirty minute endurance rating.

3.21.2.4 Toxicity

Materials used shall comply with Section 16.2.4 of Attachment 7 to the Concession Agreement, Part C or the following standard.

- SMP 800C
- NF X 70-100
- BS 6853 Annex B2

3.21.2.5 Electrical Fire Safety

Electrical equipment, wiring and apparatus shall conform to the standards of NFPA 130, Section 8 or equivalent.

3.21.2.6 Fire and Toxicity Protection Program Report

The Contractor shall submit to Company a report (the ***Fire and Toxicity Protection Program Report***) that shall demonstrate compliance with the applicable requirements and standards as set forth in Section 16.2 of Attachment 7 to the Concession Agreement, Part C. The report shall include the sections detailed in Section 16.2.6 (a) through (e) of Attachment 7 to the Concession Agreement, Part C: [CDRL #7C-19]

3.21.3 Software and Software Systems

The Contractor shall be responsible for the overall design and for the integration of individual Rolling Stock software systems into the complete system in accordance with the requirements of this Section 16.3 of Attachment 7 to the Concession Agreement, Part C.

3.21.3.1 Software Systems Requirements

Subsystem software shall be designed and documented by the supplier of such software and Contractor shall review the designs and test the completed systems. Contractor's process for software review and testing shall be reviewed and audited by the Company.

3.21.3.2 Software Documentation

(a) Software documentation shall be in accordance with IEEE Standard P1558-2004, Standard for Software Documentation for Rail Equipment and Systems (IEEE 1558) and with the additional requirements set forth within Section 16 of Attachment 7 to the Concession Agreement, Part C. Software documentation shall use a type-5 system as defined within the IEEE 1558 standard.

(b) The Contractor shall maintain the software documentation as detailed in Section 16.3.2 (b) (i) through (iii) of Attachment 7 to the Concession Agreement, Part C.

(c) The scope of the documents maintained by the Contractor in accordance with paragraph (b) shall cover the entire software development for the Cars. The SVVP, as detailed in Section 3.21.3.5 below, shall describe the integration of the subsystems to cover the requirements of the entire system.

(d) A Software Verification and Validation Report (*SVVR*) shall be prepared with each software release documenting the Contractor's verification and validation activities as required by Section16.3.1 of Attachment 7 to the Concession Agreement, Part C.

3.21.3.3 Not Used.

3.21.3.4 Software Project Management Plan

The SPMP shall comply with Section 16.3.3 of Attachment 7 to the Concession Agreement, Part C.

3.21.3.5 Software Verification and Validation Plan

The SVVP shall comply with Section 16.3.4 of Attachment 7 to the Concession Agreement, Part C.

3.21.3.6 Software Requirements Traceability Matrix

The SRTM shall comply with Section 16.3.5 of Attachment 7 to the Concession Agreement, Part C.

3.21.3.7 Software Configuration Control Plan

The Contractor shall develop, implement and maintain a plan (a ***Software Configuration Control Plan***) for tracking software changes relative to Cars, portable test equipment, and bench test equipment. The Contractor shall include in the plan a database management system capable of maintaining the history of all software and status changes making it possible to determine which versions currently resides in which equipment, on which Cars, and also which versions were used in the past. The database management system shall be capable of generating various reports showing the configuration of a typical Car system in terms of software history and status, and also the configuration of the fleet in a report capable of determining the exact software configuration of a particular operating train on a Car-by-Car basis.

4 VERIFICATION AND DEMONSTRATION

The Contractor shall be responsible for the Rolling Stock related activities identified in Attachment 7 to the Concession Agreement, Part D:

4.1 VERIFICATION PROGRAM

4.1.1 Verification Requirements

(a) The Contractor is responsible for comprehensively demonstrating compliance of the Work with the requirements of the Contract. Verification shall encompass design, procurement, fabrication, assembly, construction, installation, inspection, integration, testing and commissioning. Verification may be achieved through analysis, inspection or testing.

(b) The Contractor may provide draft versions of any verification and testing submittal defined in Attachment 7 to the Concession Agreement, Part D for review at Periodic Design Review Meetings prior to submission of the final forms of such submittal to the Company.

4.1.2 Contract Compliance Verification Matrix

(a) The Contractor shall develop a matrix (the Contract Compliance Verification Matrix or CCVM) identifying how each technical requirement of the Contract is to be met and allocating a verification action to each such requirement. Each verification action shall include a forecast completion date, which shall be consistent with dates for the associated activity shown on the Company's Project A Schedule, Exhibit 1 . Where appropriate, cross reference shall be made to the System Testing and Commissioning Plan and to other relevant Contractor Design Submittals. The Contractor shall submit the CCVM to the Company for approval. [CDRL #7D-01]

(b) The Contractor shall maintain the approved CCVM throughout the Contract duration, shall record completion and approval of each verification action in the CCVM, and shall submit updates to the Company for review not less than every three months until each requirement has been verified. [CDRL #7D-02]

4.1.3 Verification through Analysis

Where the Contractor provides analysis as a means of verifying compliance with technical requirements, the Contractor shall specifically identify the submittal as "Analysis of Requirements Compliance" and include the applicable CCVM cross reference and Contract reference on the submittal title page. The Contractor shall ensure that each verification analysis is verified and agreed by a reviewer with appropriate expertise that is independent of the design entity preparing such verification analysis. The Company reserves the right to request and review analyses identified as the means of compliance verification.

4.1.4 Verification through Inspection

Where the Contractor intends to provide inspections as a means of verifying compliance with technical requirements, the Contractor shall prepare specific inspection procedures that identify the applicable CCVM cross reference, and the methodology through which the inspection demonstrates compliance. The Contractor shall conduct each inspection in accordance with the inspection procedures and shall document the results in an inspection report.

The Contractor shall ensure that each inspection procedure and inspection report is reviewed and accepted by a suitably qualified reviewer that is independent of the design entity preparing such verification analysis. The Company reserves the right to request and review any inspection procedures and reports identified as the means of compliance verification.

4.2 SYSTEM TESTING AND COMMISSIONING

The Contractor is responsible for carrying out necessary testing and commissioning of Rolling Stock project elements required to ensure and demonstrate that the Cars are safe, functional and reliable, and meet all Revenue Service Commencement Requirements related to system performance, prior to the relevant Revenue Service Commencement Date. Specific tests identified in this document shall be

included in the Company's test program. Testing shall also include those tests specifically required by Relevant Authorities.

4.2.1 System Testing and Commissioning Plan

(a) The Contractor shall develop a plan (the System Testing and Commissioning Plan), which details all tests to be carried out, the scheduled test date and location of the test, the function to be tested and/or requirement to be demonstrated, and the party responsible for the testing. [CDRL #7D-03]

(b) The System Testing and Commissioning Plan shall differentiate between:

(i) testing to demonstrate functional design compliance and performance verification (qualification tests);
(ii) testing for acceptance of each component, subsystem and system (acceptance tests);
(iii) testing of integrated systems (integration tests); and
(iv) demonstration of the performance of the Rolling Stock (demonstration tests).

(c) The System Testing and Commissioning Plan shall detail how the Contractor will manage the test program and the resources to be applied during testing and commissioning. It shall include:

(i) the program, sequence and schedule for all testing and commissioning;
(ii) control and monitoring processes for the progression of the testing and commissioning;
(iii) methods for reporting on completed tests;
(iv) testing documentation control procedures;
(v) the process for managing non-conformances and re-testing;
(vi) management of safety during testing and commissioning, including training of staff and contingency provisions; and

(vii) the formal procedures for the transition from commissioning personnel to operations and maintenance personnel and fulfilling the Revenue Service Commencement Requirements.

(d) The Contractor shall reissue the System Testing and Commissioning Plan monthly once testing has begun, updating information as test scheduling is confirmed and tests are carried out, and annotating which tests the Company and the Owner will witness. [CDRL #7D-04]

(e) The Contractor shall, on completion of acceptance testing of the final Car, or three months prior to start of revenue service whichever is sooner, submit to the Company for approval a final version of the System Testing and Commissioning Plan, recording all testing carried out, functions and performance demonstrated, reports produced and approved; and third party approvals, permits, certificates and authorizations for the start of revenue service. [CDRL #7D-05]

4.2.2 Rolling Stock Testing and Initial Acceptance

(a) The Contractor shall perform a comprehensive testing program to demonstrate that the Rolling Stock meets the requirements of applicable Law, of this Scope of Work. The test program shall include Owner-required tests as set forth in Section 2.5 of Attachment 7 to the Concession Agreement, Part D, and Attachment 7 to the Concession Agreement, Part C (*Rolling Stock Requirements*). As applicable, test submittals shall be provided for each unique Car type.

(b) For testing specifically required in the Contract, the Company may accept previous test reports in lieu of new testing in its sole discretion, where the Contractor can demonstrate to the Company that the equipment testing and the service application (including environmental conditions) were the same as for this Work. At its discretion, the Contractor may consider the application of previous test reports and/or service history records in lieu of other new testing, but only where equivalency of such previous testing means the definition stated above.

(c) Rolling Stock test procedures, the conduct of Rolling Stock tests and the pre-revenue service acceptance test plan shall be in accordance with 49 CFR 238.111.

(d) The Contractor shall develop, during the design phase, a Rolling Stock test matrix listing tests to be performed within the Rolling Stock test program, which shall form part of the overall System Testing and Commissioning Plan (see Section 2.1 (*System Testing and Commissioning Plan*) of this Part). The test matrix shall be updated regularly to accurately capture any changes to program scope or direction.

(e) Upon arrival at Site.

(i) A visual inspection and water tightness test shall be conducted by the Contractor in the presence of the Company upon arrival of each Car.

(ii) Selected static and limited dynamic tests, previously conducted at the Contractor's final assembly site will be repeated as verification of the general condition of the Car. Additional testing may be conducted in response to anomalies in the visual or functional condition of the Car.

(iii) Limited system integration tests shall be conducted including train control (successful communication, train to wayside and reverse), communications upload and response characteristics, CCTV functionality, etc.

Testing at the Company's facilities may be limited by available track and/or systems installations. Testing shall be adjusted in response to such conditions as encountered at the time of arrival of each Car.

The Company shall make reasonable efforts to ensure that the initial acceptance of each Car shall not be negatively impacted by the lack of availability of track, facilities or systems.

At the successful conclusion of the above testing regime, a Car will be Initially Accepted (Initial Acceptance) by the Company, and the transfer of title from the Car builder to Owner will be authorized.

4.2.3 Test Procedures and Reports

(a) Each test procedure shall clearly identify the function and performance that it is intended to verify (including pass/fail criteria) and shall include appropriate cross references to applicable section(s) and items within the CCVM, the System Testing and Commissioning Plan and the Agreement.

(b) No test shall be performed by any Subcontractor without a test procedure that has been approved by the Contractor. Testing performed by Subcontractors shall be witnessed by the Contractor.

(c) Tests to be performed by the Contractor shall not be performed without a test procedure that has been approved by the Company. [CDRL #7D-06]

(d) Any test found to have been performed without an approved test procedure shall be repeated in the presence of a Company witness, and the associated test report shall be reissued.

(e) Test reports shall be prepared for all tests performed, regardless of the outcome. The Contractor shall submit reports of integration tests and demonstration tests, as defined in Section 2.1 above.

4.3 TEST WITNESSING

4.3.1 Company Test Witnessing

In response to the submittal of any System Testing and Commissioning Plan, the Company will provide the Contractor with its anticipated plan to witness certain testing and to require review of certain test procedures and test reports. The Company reserves the right to witness any and/or all tests, and to require submission of any and/or all test procedures and reports. [CDRL #7D-07]

4.3.2 Owner Test Witnessing

In response to the submittal of any System Testing and Commissioning Plan, the Owner will provide the Company with its anticipated plan to witness certain testing and to require review of certain test

procedures and test reports. The Company shall advise the Contractor of these plans. The Owner reserves the right to witness any and/or all tests, and to require submission of any and/or all test procedures and reports.

4.3.3 Owner-Required Testing for Rolling Stock

The Contractor shall incorporate the test titles as detailed in Section 2.5 (a) through (e) of Attachment 7 to the Concession Agreement, Part D into the Contractor's test program for the Rolling Stock unless an alternative verification method is agreed by the Company. The Contractor shall define the scope and magnitude of these tests in order to fully demonstrate compliance with the requirements of this Scope of Work. Expectations for these tests are included in *Attachment 7 to the Concession Agreement, Part D.*

4.4 INTEGRATION TESTING

The Contractor shall support the Company in the performance of integration testing to demonstrate the correct operation and full functionality and performance of subsystems and systems together, in accordance with an approved integrated testing program. The integrated testing program shall be consistent with the System Testing and Commissioning Plan and shall include prerequisite testing for the subsequent system performance demonstration program.

4.4.1 Integrated Testing Program Plan

(a) The Contractor shall submit to the Company a plan (the Integrated Testing Program Plan – Rolling Stock), which shall include: [CDRL #7D-08]

(i) a list of all integration tests to be performed;

(ii) the interfaces and functions that are to be proven by the integration tests;

(iii) requirements conformance to be proven, cross-referenced to applicable parts of the Agreement and the CCVM; and

(iv) the integrated testing schedule.

(b) Integrated testing shall, at a minimum, include testing of:

(i) Rolling Stock gauging and interfacing with the track geometry and clearances;

(ii) Rolling Stock interface testing with CRMF shop equipment;

(iii) Rolling Stock signal, and train control system interface operation, including ATC and PTC;

(iv) Rolling Stock communications systems operation;

(v) Rolling Stock passenger information systems; and

(vi) Rolling Stock emergency and safety related equipment interfaces, such as overspeed protection, bumping posts, emergency egress methodologies;

(vii) Rolling Stock interface testing with TES

4.4.2 Integrated Testing Procedures and Reports

(a) At least 3 months prior to the scheduled date of each integrated test, the Contractor shall submit the integrated testing procedures and test report forms to the Company. [CDRL #7D-09]

(b) Test reports shall be submitted within 21 days after completion of each integrated test. [CDRL #7D-10]

4.5 SYSTEM PERFORMANCE DEMONSTRATION

4.5.1 System Performance Demonstration Plan and Procedures

(a) The Contractor shall participate and support the Company in the preparation of a plan (the ***System Performance Demonstration Plan***) demonstrating the approach to performing the system performance demonstration program and verifying the readiness of the Commuter Rail System, including Rolling Stock, and operating and maintenance personnel for revenue service. [CDRL #7D-11]

(b) Following approval of the System Performance Demonstration Plan, the Contractor shall assist in the development of procedures (the ***Performance Demonstration Testing Procedures***), which shall include the processes for performing the tests, recording and analyzing the performance data, and reporting on the readiness of the Commuter Rail System, including Rolling Stock,

(c) Prior to each Revenue Service Commencement Date, the Contractor shall assist in the performance of demonstration tests on the Rolling Stock in accordance with the submitted procedures. These tests will be witnessed by the Owner.

4.5.2 System Performance Demonstration Program

4.5.2.1 Goals of the System Performance Demonstration

(a) As the culmination of the integrated test program, the Contractor shall support the Company in the performance of a comprehensive demonstration (the ***System Performance Demonstration***), the objective of which is to test the complete integrated Commuter Rail Network, including subsystems, operating personnel and operating procedures; and to confirm readiness for entry into revenue service. The System Performance Demonstration shall include:

(i) testing of anticipated normal and abnormal revenue service operations;
(ii) simulation of failure and emergency scenarios, to test failure management procedures and personnel readiness for revenue operations; and
(iii) testing required by Project Third Parties and Relevant Authorities.

(b) The System Performance Demonstration shall demonstrate the readiness of the Company to provide safe and reliable passenger service to the required service pattern, and ensure that the Company-operated Components, such as the Rolling Stock, are achieving sufficient reliability for sustainable service by:

(i) familiarizing the operating and maintenance staff with the operation of the integrated system;
(ii) exercising and validating the operating schedules and procedures; and

(iii) exercising and confirming the operating reliability of the subsystems in simulated operating scenarios.

4.5.2.2 Features of the System Performance Demonstration

(a) Testing shall be conducted in a progressive manner, incrementally increasing the size and complexity of the network being operated until an entire Commuter Rail Project is under test.

(b) The Company's System Performance Demonstration Plan shall identify the manner in which testing will progress and shall at a minimum demonstrate the following:

(i) the ability to achieve maintenance plans and schedules through a maintainability demonstration to be conducted by the O&M Contractor in conjunction with the Contractor;
(ii) the ability to dispatch trains from the CRMF as required for scheduled morning roll-out;
(iii) the operation of limited line segments (in various locations) that includes stations, grade crossings and service turnback;
(iv) the operation of any single track line segments at scheduled peak period headways, including minor disturbances/delays to some trains;
(v) the operation of each remote terminal station and DUS, with trains arriving and departing according to scheduled peak period headways, including minor disturbances/delays to some trains;
(vi) the simulated service operation of each entire Commuter Rail Project, at off-peak and then at peak period headways;
(vii) the performance under abnormal and emergency scenarios, included rescue of a failed train, partial loss of power supply, and recovery of service after a major disruption; and
(viii) the simulated service operation of both corridors concurrently at full scheduled service levels.

(c) For any corridor already in revenue service, testing of simulated service on any other corridor shall not interfere with the Company's ability to maintain required service standards on such corridor in revenue service.

Contractor support shall be provided as required for this testing.

4.5.2.3 Performance Duration

(a) The full System Performance Demonstration for each Commuter Rail Project shall last at least 21 days, with a minimum of 10 consecutive hours of operation each day, including transitions between peak and off-peak service that represent a full weekday operation.

(b) The System Performance Demonstration shall continue until the relevant Commuter Rail Project has achieved the Availability Ratio required by the Revenue Service Commencement Requirements, as defined in the Contract.

(c) Concurrently with the System Performance Demonstration or as an integrated part of the System Performance Demonstration, the Company shall operate the relevant Commuter Rail Project for 20 hours each day for 14 consecutive days in order to acclimatize personnel and communities to normal service levels.

4.5.2.4 System Performance Demonstration Reports

(a) During the System Performance Demonstration, the Contractor shall prepare and submit to the Company on a daily basis status reports on the progress of the demonstration testing relating to the Rolling Stock and a summary of each incident that occurs causing a reduction in the performance metrics. [CDRL #7D-13]

(b) During the System Performance Demonstration, the Contractor shall prepare and submit to the Company status reports on a weekly basis that provide a complete summary of testing progress, identify

trends in performance issues and describing corrective actions that are being implemented to improve system performance. [CDRL #7D-14]

(c) Once the System Performance Demonstration has successfully met the required Availability Ratio as defined in the Contract, a complete Rolling Stock test report shall be prepared and submitted within 21 days of test completion. [CDRL #7D-15]

(d) If the threshold requirement has not been met after the initial 21 days of the System Performance Demonstration, the Contractor shall prepare and submit a draft report, identifying the contributing reasons relating to the Rolling Stock for failing to achieve the milestone and a corrective action plan.

4.5.2.5 Rolling Stock Distance Accumulation

(a) Revenue service may not begin on any Commuter Rail Service until the Rolling Stock has cumulatively run a minimum of 25,000 Car miles. No Car shall enter revenue service until such Car has accumulated a minimum of 1,000 miles of test running.

(b) The Contractor shall prepare a report (a ***Distance Accumulation Report***) recording the mileage run by Car, by day and by Commuter Rail Project, for each Car once the threshold value set forth in paragraph (a) is attained for that Car and prior to that Car entering revenue service. Where Cars are semi-permanently coupled into minimum operable units, documentation may be produced at such unit level.

5 RELIABILITY, AVAILABILITY, MAINTAINABILITY AND SAFETY (RAMS)

The Contractor shall develop and deliver the Cars as needed to meet expected levels of reliability, availability, maintainability, and safety of modern EMU Rolling Stock. The approaches, processes, and methods used by the Contractor to fulfill this requirement shall be proven to be effective in meeting these goals. The Contractor shall work with the Company to improve reliability, maintainability, availability, and safety, and reduce risk to reasonable levels as expected in the operation and maintenance of modern EMU Rolling Stock. The Contractor shall identify the method for identifying and reporting performance in these areas unless otherwise specified within this section.

The Contractor shall develop a Rolling Stock Reliability Management Plan, identifying the target Car reliability, providing sub-system by sub-system performance analysis to demonstrate that the reliability target is achievable.

The Contractor shall provide an FMECA study of the Car

The Contractor shall identify a single person responsible for the RAMS relating to this project.

The following Reliability Definitions shall apply for all matters pertaining to RAMS for the Rolling Stock.

Assembly	A portion of a Subsystem defined as the design evolves, which consists of Parts, Components or both.
Component	A portion of an Assembly not amenable to further disassembly by the Company for maintenance or repair by the Company
Failure	An improper condition of Items which requires maintenance or replacement to restore affected Items to their normal operating condition
Failure, Component	The failure of any Part, Component, Assembly, System, Subsystem, or Car to function or perform in accordance with this Specification, or the requirement for any such Item to undergo unscheduled maintenance.
Failure, Independent	An Independent Failure is a Component Failure, not caused by the failure of another Component, Assembly, Subsystem, System or Car
Failure, Indeterminate	A temporary classification of failure condition, pending determination of proper classification by the Failure Review Board (FRB).
Failure Mode	The consequence of the mechanism through which the failure occurs, i.e., short, open, seized, noisy, etc
Failure, Non-Relevant	Any failure or condition of an Item which meets one or more of the following criteria: • A failure caused by malfunctions of other equipment. • A failure caused by human error, except as noted below for Relevant Failures. • A failure caused by accidents not associated with the normal operation of the item, such as collision or striking a foreign object on the right-of-way. • A failure caused by a delay to another consist. • A failure caused by operating the item outside of design or environmental specification limits. • A failure caused by malfunction of Components not furnished, designed, manufactured, modified, or installed by the Contractor, Subcontractor, or Manufacturer under the provisions of these Specifications, or which is shown by Contractor's analysis and Failure Review Board concurrence to be out of Contractor's scope of responsibilities under these Specifications.

	• Consumable items requiring replacement during specified preventive maintenance or which should have been replaced during a previously scheduled preventive maintenance interval and were not. • A failure which is a recurrence of one that was thought to have been corrected. (The first failure shall be classified as relevant, but not the recurring one, if the failure occurs within 400 Car operating miles of the initial failure.) • Reported failures for which, upon final review, insufficient or inadequate documentation is obtained to determine failure relevancy. • A failure caused by vandalism or abuse to the Car.
Failure, Pattern	The occurrence of three or more failures of the same or identically replaceable Items (i.e. the same part number) within the lesser of 10,000 fleet hours of operation or 400,000 fleet miles of travel, in identical applications or functional modes
Failure Rate	A measure of the mean rate at which Items from the same application population fail. The failure rate is determined by dividing the total number of failures by the total number of life units expended by that application population during a particular measurement interval. The life units used within the Company are "miles". Also, categories of failure, such as Component, Service and Train Delay, which relate to different failure consequences, are used to quantify different failure counts and express different types of failure rates that are specified in this Scope of Work
Failure, Relevant	A Relevant Failure of an Item is an independent failure plus any dependent failures caused thereby which results in loss of function of that Item caused by any of the following: · • A fault in the Item while operating within its design and environmental specification limits. • Improper operation, maintenance, or testing of the Item as a result of Contractor supplied documentation or other Contractor failure to meet its contractual obligations. • Item degradation, including failure, defects, or non-conformance discovered during recommended preventive maintenance interval (excluding consumable items), which required removal and replacement, corrective maintenance, and/or disposition. • Loss of function of any item which cannot be duplicated when investigated and, therefore, cannot be corrected by a specific maintenance action ("no fault found" or "no trouble found"). • Failures or failure indications which require operator/maintenance technician corrective action prior to return of that item to a functioning state. • Consumable items requiring replacement for reasons other than normal fleet wear/aging. • Software failures which manifest themselves in an equipment failure or in an indication of failure, resulting in loss of function of an item in accordance with the Specification. • Pattern failures that have been so identified by the Failure Review Board

	(FRB), and for which corrective action has been implemented, will be assessed as one relevant failure. The corrective action must be proven and effective.
Failure, Service	A Service Failure is a Component Failure that meets any one of the following additional criteria i. The Car is taken out of service at the start of or en route to its destination due to a problem or failure that has occurred resulting in an unsafe condition, or due to a problem or failure which prevents the Car from completing its Mission. ii. The Car, after beginning its Mission, continues to its final destination but a problem or failure has occurred that makes it unacceptable for continued service due to the need to perform maintenance or repairs. iii. The Car is unacceptable for service as determined by the Company due to a problem or failure that is detected prior to the initiation of the Mission. iv. The Car is removed from service for an unscheduled repair which, if left unattended, could result in fine or penalty being assessed in accordance with 49 CFR 200 to 399.
Failure, Train Delay	A Train Delay Failure is a Component Failure that causes a train delay of more than 4 minutes, 59 seconds in completing a mission
Item	A non-specific term used to denote any portion of a Vehicle or Car, including software. The Item Levels used by the Company from the simplest to the more complex are as follows: Part, Component, Assembly, Subsystem, System, and Car
Mean Distance Between Component Failure (MDBCF)	A measure of the reliability of an Item expressed as the mean operating mileage traveled between all Relevant Component Failures. MDBCF (in miles) is the reciprocal of the Component Failure rate, λc, for the Item, measured in Component Failures per mile. See Table 204-1
Mean Distance Between Service Failures (MDBSF)	A measure of the reliability of an Item expressed as the mean operating mileage traveled between all Relevant Service Failures. MDBSF (in miles) is the reciprocal of the Service Failure rate, λs, for the Item, measured in Service Failures per mile. See Table 204-1
Mean Distance Between Failures (MDBF)	A measure of the reliability of a Car expressed as the mean operating distance mileage traveled between all Relevant Train Delay Failures. MDBF (in miles) is the reciprocal of the Train Delay Failure rate, λtd, for the Item in Train Delay Failures per mile. See Table 204-1
Mission	A one-way trip from the beginning terminal to the destination terminal as defined by the Company.
Part	Any portion of a Component not amenable to further disassembly by the Contractor and its Subcontractors or Suppliers for repair purposes
Subsystem	A portion of a System defined in the Technical Specification and refined in the design process consisting of Parts, Components and/or Assemblies
System	One of the Systems specified in Attachment 7 to the Concession Agreement, Part C

5.1 RELIABILITY

Reliability of the Rolling Stock shall, meet the overall vehicle reliability requirements outlined herein. The Contractor shall provide reliability monitoring of Rolling Stock performance above and beyond the initial performance demonstration, implementing reliability improvement programme as required to identify and rectify root causes of the vehicles not achieving the required performance in accordance with Part III Section 2.22.2. This activity shall continue until performance exceeds the reliability target for a sustained period of 3 months. A monthly reliability performance report shall be produced by the Contractor.

The following table gives the reliability requirement for the Rolling Stock. Reliability shall be based on Section 4 in General Description of the Rolling Stock Design Concept Approach (C.12 Technical Approach Rolling Stock Sections A-S Technical Description of the Rolling Stock Design Concept and Technical Approach) and the proposed overall vehicle reliability target is shown in the table below. For the purposes of determining if Contractor has exceeded the overall reliability target, only the overall vehicle reliability target shall be used, not the subsystem reliability values.

SYSTEM / SUBSYSTEM	MDBSF REQUIREMENT (MILES)	MDBF REQUIREMENT (MILES)
Door Control System & Doors	1,400,000	2,800,000
Communications Systems / Passenger Information Signs	2,500,000	N/A
Event Recorder	4,000,000	N/A
Monitoring and Diagnostic System	5,000,000	N/A
HVAC System	750,000	N/A
Primary Power Distribution and Auxiliary Power System, Low Voltage System, Trainlines and Train and Car Control, and Pantograph Current Collector	1,000,000	N/A
Propulsion System and Adhesion Management	500,000	2,500,000
ATC/PTC Systems	2,500,000	2,500,000
Friction Brake System, Compressed Air System, Parking Brake System, Adhesion Management Dump Valves	1,000,000	1,000,000
carbody	3,000,000	N/A
Interior Furnishing / Finishes	3,000,000	N/A

SYSTEM / SUBSYSTEM	MDBSF REQUIREMENT (MILES)	MDBF REQUIREMENT (MILES)
Lighting	4,500,000	N/A
Coupler (Mechanical)	10,000,000	10,000,000
Truck Assemblies	1,500,000	N/A
Overall Vehicle	**90,000**	**140,000**

5.2 AVAILABILITY

The Contractor shall work with the Company to develop a target Availability requirement for the vehicles, based on system operations, vehicle system reliability, vehicle maintainability, including maximum component / sub system removal and replacement times, and the maintenance programme as recommended by the Contractor.

The Contractor and the Company shall jointly produce availability reports for the accepted fleet as part of the service performance report.

5.3 MAINTAINABILITY

For unaltered aspects of the vehicle design, the Contractor shall supply maintainability information relating to the design, including details relating to accessibility for maintenance activities, defining the planned maximum maintenance downtime for removal and refit activities. In addition, the Contractor shall provide test reports relating to demonstration of these activities.

For any and all new aspects of vehicle design developed specifically for this project, the Contractor shall incorporate focus on maintainability, including accessibility for maintenance activities, defining and demonstrating the planned maximum maintenance downtime. Maintainability will form part of the design review process, and be reported upon at design review meetings. Demonstration of maintainability will be performed at First Article Inspections, mock up reviews if applicable, and pilot Car testing.

5.4 SYSTEM SAFETY

System safety shall be based on Section 4 in General Description of the Rolling Stock Design Concept Approach (A159-TP-A-01001) and general approach is as follows;:

The Contractor places great emphasis on achieving the safety and quality goals required to satisfy our customers' and applicable regulatory agencies' safety requirements, and take great pride in our reputation for the production of high quality, safe and reliable rail transportation vehicles.

The primary regulations, requirements and documents relevant to System Safety include:

- 49 CFR Part 238, Subpart B
- Fire Safety
- Train electronic hardware and software safety
- American Public Transportation Association, Manual for the Development of System Safety Program Plans for Commuter Railroads
- MIL-STD-882C - Section 4 and all tasks within Sections 100 and 200 of that document
- CENELEC Standard EN50128, Railway Applications: Software for Railway Control and Protection Systems, especially sections 8 through 17

System Safety is an integral part of Contractor's overall System Assurance Program covering Safety, Reliability, Maintainability, and Quality Assurance. The System Assurance Program Plan (SAPP) is the governing mechanism for Contractor's Systems Assurance Program. The SAPP is intended to define both the management and technical activities, defined in the formal Safety, Reliability, Maintainability and Quality Assurance Plans, to be performed during conceptual analyses, design, installation and testing of the new EMU vehicles. The SAPP and its integrated subordinate plans provide guidance for demonstrating that the EMU vehicles designed and manufactured by Contractor will be compliant with the System Assurance and safety requirements of the Specification.

The SAPP covers the individual cars, and the individual subsystems and equipment, and includes both hardware and software aspects.

This SAPP governs the activities and tasks of all staff of Contractor, subcontractors, and suppliers in the performance of systems assurance tasks. Further, it will govern the activities and tasks for the duration of the project and the warranty period.

The SAPP will serve the objective of providing RTD and HR Project Managers with a complete statement of the planned System Assurance Program for the new vehicles. This information is required for successful planning of manpower and material resources, as well as to define the guidelines for assuring and demonstrating that the EMU's safety, reliability, maintainability and quality requirements meet the overall project objectives. Another objective of the SAPP is to provide the integrated structure for the individual elements of the SAPP.

Contractor's SAPP generally contains the following items.

Section	Title	Contents
1	Introduction	Provides the context for the SAPP, i.e., purpose, objectives, standards and scope
2	Glossary of Terms	Provides a list of definitions and abbreviation
3	Scope	Provide all applicable tasks based on MIL-STD-882C
4	Safety Requirements	Summarizes the safety requirements
3	Project Organization	Provides senior level organization chart and responsibilities
4	Systems Assurance Management	Provides systems assurance organization chart and responsibilities
5	Methodology and Approach	Describes the activities that are common to the different assurance programs and the means for integrating these different aspects
6	Safety analysis	States the objectives, metrics and methodology for ensuring system safety and lists the contract deliverables of the safety program It describes in detail tasks and activities of system safety management and system safety engineering required to identify, evaluate, and eliminate hazards, or reduces the associated risk to an acceptable level throughout the system life cycle.

The process of identification and review and for safety critical items occurs on several fronts.

Safety critical items are initially identified by the design team based on the requirements of the Technical Specification and applicable regulatory requirements, such as 49 CFR Part 238. Specific safety requirements are also indicated on documentation handled through Contractor's own corporate procedures.

Such documentation includes supplier/subcontractor Requests for Quotation, drawings, and technical descriptions. Where specific certifications are required, these are specified and reviewed for compliance during supplier qualification, first article inspection, qualification testing, and/or receiving inspection, or other first opportunity, as required. When analysis and review activities are required, Contractor shall obtain the needed data and conduct review and analysis for acceptability prior to submittal to the Company.

Hazard identification and initial risk classification is to be accomplished by means of a variety of hazard analysis techniques and sessions, including Preliminary Hazard Analyses (PHA), Subsystem Hazard Analysis (SSHA), System Hazard Analysis (SHA), Operating & Support Hazard Analysis (O&SHA), Hazard Log and FMECA (Failure Mode and Effectiveness Analysis).

6.0 MATERIAL, PERMANENT EQUIPMENT, OR PERMANENT WORK SITE SERVICES FURNISHED BY COMPANY

Company will furnish or cause to be furnished to Contractor, without cost to Contractor, the material, permanent equipment or permanent Work Site services set forth herein, for or in connection with, performance of the Work. Unless specifically set forth herein, Contractor shall provide all other material, permanent equipment or permanent Work Site services required for the Work.

Material and Permanent Equipment

6.1 None provided by Company.

6.2 Permits

Notwithstanding the Article entitled "Permits and Licenses" in Part III – General Terms, Company will furnish the permits listed below:

All permits required for performance of the Work at the Work Site will be arranged by Company, provided Contractor shall in a timely manner prepare all applications for such permits and submit them to Company.

All inspection of the Work required by governmental agencies will be arranged by Company. Contractor shall request such inspection through Company only after the Work is ready for inspection.

7.0 TEMPORARY SITE FACILITIES AND UTILITIES

7.1 Furnished by Company

A site office, or location for a temporary site office will be provided at the CRMF for delivery, acceptance testing, commissioning, and warranty support personnel. These shall

be furnished with electrical power. Water, washroom facilities, telephone and internet connections may or may not be provided.

7.2 Furnished by Contractor

7.2.1 Material Handling and Rigging

Contractor shall provide and operate all cranes and other necessary equipment for handling, hauling, unloading and receiving Contractor supplied materials, tools and equipment.

7.2.2 Personnel Protective Equipment

7.2.3 Transportation facilities on and off site. Only Contractor's company vehicles, as approved by Company, will be allowed on the Work Site (CRMF). It should be noted that parking facilities are not guaranteed.

7.2.4 First aid facilities.

8.0 CAPITAL SPARES, TEST EQUIPMENT, SPECIAL TOOLS AND INVENTORY

The Contractor shall furnish the following capital spares, test equipment, and special tools. If a Rolling Stock Option is exercised, the proportional number of capital spares will be provided. Selected special tools and test equipment shall also be required depending on the nature of the tools/equipment and its intended usage.

8.1 Capital Spares

A minimum of 2 Car sets of all replaceable/rebuildable interior body units will be provided as capital spares deliverable to the Site before the delivery of the last production Car of the Base

Order. This provision will include but not be limited to all interior body panels and fixtures, seats, window assemblies, windshields assemblies door leafs, end doors, and other body parts as may be damaged and require replacement.

A minimum of 3 Car sets of all replaceable/rebuildable mechanical/electrical units will be provided as capital spares deliverable to the Site before the delivery of the last production Car of the Base Order. This provision will include but not be limited to fully dressed (built-up, ready to install) trucks, traction motors, air compressors, air conditioning compressors, air conditioning condensers, complete air conditioning units (including frame), propulsion equipment, auxiliary power equipment, train control equipment, communications/radio equipment, safety equipment, master controller, operators "desk" equipment, door operators and other major mechanical, electrical, and electromechanical equipment.

A minimum of 4 Car sets of couplers and draw bar assemblies will be provided as capital spares deliverable to the Site before the delivery of the last production Car of the Base Order. In addition, 4 compromise couplers will be provided.

8.2 Special Tools

All tools, excluding tools commonly available from standard suppliers such as Granger and/or Snap-On that are required or desirable for the effective/efficient conduct of maintenance and repair of the vehicles will be provided as a "special tool" in quantities necessary to support the level and intensity of vehicle maintenance anticipated. This requirement extends through mechanical, electrical, electromechanical, and electronic tools, including software products. These tools will be provided within 60 days of the delivery of the first production Car of the Base Order.

8.3 Portable and Bench Test Equipment

All test equipment required or desirable for the conduct of maintenance and repair activities of the Car will be provided in sufficient quantity to adequately support the maintenance operation. All such test equipment to be provided by the Contractor regardless of the equipment's availability from other common sources. Portable test equipment (Two of each kind, including instruction manuals, cables, connectors and associated software), shall be provided for all on-board systems. All portable test equipment shall be delivered within 30 days of the delivery of the first production Car of the Base Order. One set of bench test equipment shall be provided for the purpose of testing, troubleshooting and calibrating all electrical, electronic, mechanical and electro-mechanical components of each vehicle subsystem. All test equipment, including bench test equipment, shall be delivered within 90 days of the delivery of the first production Car of the Base Order.

8.4 Inventory

The Contractor will provide an initial stock of spare parts and consumables within 60 days of delivery of the first production Car. The initial stocking levels of the Inventory will be at a minimum level of one year's anticipated usage of all spare parts and consumables, over and above the Capital Spares. The inventory will be maintained by the Contractor at approximately this level for the duration of the three (3) year General Warranty period. At the conclusion of the General Warranty period, the Contractor will adjust the on hand quantity of all parts and consumables to a level equal to actual usage for the third year of service. At such time, responsibility for maintaining the inventory will pass to the O&M contractor. The Inventory shall be included in the price of the base order of Cars. "Warranty stock" maintained by the Contractor will be separate and distinct from the inventory. Any residual warranty stock will be made available for purchase by the Company on a part-by-part basis, at the Company's sole option, at an agreed upon price.

Subsequent purchases of parts and consumables will be priced at "best-customer" pricing, and that best-customer pricing will be made available to the company until the last vehicle is retired from service.

9.0 PERFORMANCE SCHEDULE AND SEQUENCE OF WORK

9.1 Contractor shall commence performance of the Work and shall complete the Work in accordance with the dates set forth in the Company's Project Schedule, Exhibit 1.

9.2 Specific milestones, interfaces and other schedule related bases of this Contract are as set forth in Exhibit One.

9.3 General scheduling, reporting and coordination requirements shall be described in Part III, General Terms and more specifically in Attachment G – Scheduling and Reporting.

9.3.1 Contractor shall submit the detailed schedule required by Section 10.1 of Part III, General Terms within thirty (30) calendar days of award of this Contract.

9.3.2 Specific scheduling and coordination requirements may include, but not necessarily be limited to the following:

A. Engineering deliverables by discipline

B. Mobilization time for manpower and equipment

C. Rolling Stock deliveries to Work Site

D. Shop fabrication

E. Incremental completion dates of major components

F. Start and completion of different segments of Work (early and late starts)

G. Any qualifying conditions of Company or Owner

H. Other as necessary

10.0 REPORTING REQUIREMENTS AND COORDINATION MEETINGS

Contractor shall promptly submit the schedules and reports set forth in Attachment G and in addition, those reports as requested by Company pursuant to the Article entitled "Scheduling, Reporting and Coordination" set forth in Part III - General Terms.

10.1 Weekly Progress Meetings

10.1.1 At the weekly progress meeting, Contractor shall submit a written report showing actual manhours expended versus planned and scheduled progress versus actual progress, giving details of Work completed in relation to the approved schedule, together with a two (2) week "look ahead" which provides details of how the Work will be completed.

10.1.2 Contractor shall attend a weekly coordination meeting together with various contractors at the Work Site. Where appropriate these can be by telephone conference.

10.1.3 The person or persons designated by Contractor to attend the meetings shall have all the required authority to make decisions and commit Contractor to solutions agreed upon during any meetings.

10.2 Other Meetings

Contractor participation in certain additional activities shall also be required. These activities shall include, but not be limited to:

10.2.1 Indoctrination and orientation of all Contractor's employees prior to commencing Work at the Work Site. (This includes the entire labor force and all new hires).

10.2.2 Weekly gang box safety meetings organized and conducted by Contractor and attended by all of Contractor's craft employees. Contractor shall be responsible for arranging and conducting these meetings with its craft employees.

10.2.3 All meetings as set forth in Section 3 of this Part I – Scope of Work.

11.0 DATA REQUIREMENTS

11.1. Company will issue specifications, drawings and other documents as set forth in Attachment B, Administrative Procedure.

11.2. Contractor shall submit all data requirements as outlined in Attachment 7 to the Concession Agreement, , Part C, and the following data to Company as part of the scope of work:

11.2.1 Two (2) copies of quality control turnover packages.

11.2.2 Two (2) copies of test reports and test certificates as set forth in Attachment E and Exhibit Two within twenty-four (24) hours for review.

11.2.3 Quality control program for review prior to commencement of Work.

11.2.4 All necessary quality control documentation as Work is completed.

11.2.5 At completion of Work, a complete report of all tests.

11.2.6 All lifting and rigging procedures for review prior to lifting and delivery of Rolling Stock.

11.3 Contractor's performance of its obligations hereunder shall not be deemed complete until Company is in receipt, on proper forms, of all technical data, as-built drawings, and other documents to be submitted to Company as part of Contractor's scope of work. Failure of Contractor to comply with the above data requirements will entitle Company to withhold any progress payment, or final payment, pending Company's receipt of all the above data without prejudice to any other remedy of Company.

11.4 Contractor shall submit all engineering data, samples, and shop drawings (herein called "data") to Company for review. Company requires ten (10) working days for review of data submitted by Contractor. Each submittal of Contractor's data shall be signed by Contractor and accompanied by a letter of transmittal containing the date of submittal, Contract number, and all pertinent information required for identifying and checking submittals. One (1) reproducible and two (2) prints shall be submitted for each drawing and any other documents larger than 11" x 17". Two (2) prints shall be submitted for documents which are 11" x 17" and smaller, and documents such as procedures and calculations shall be 8½" x 11" size.

11.4.1 All drawings and specifications shall be in imperial units and the English language.

11.4.2 Drawings reviewed by Company will be returned to Contractor under the following status:

A. Proceed = "A"

B. Revise and resubmit. Work may proceed subject to incorporation of changes indicated = "B"

C. Do not proceed. Change as noted and resubmit = "C"

D. Data accepted for information only = "IO"

11.4.3 Although Work may proceed on receipt of drawing with a Code "B" notation, Contractor must incorporate the changes indicated, resubmit and obtain Code "A" notation before release of materials or equipment for shipment can be approved by Company. Returned copies of drawings with Code "B" and "C" shall be resubmitted not later than ten (10) days after the date of transmittal by Contractor of such copies of such drawings.

11.5 Contractor shall furnish to Company reproducible or electronic drawings revised by Contractor to show "as-built" information. Contractor's revisions shall show details of those locations where the Work performed by Contractor was at variance with the details shown on the drawings (either furnished by Company or furnished by Contractor and reviewed by Company). Contractor's submittal to Company of such "as-built" drawings shall be made on a continuous basis as the Work proceeds, but in all cases prior to the date of Notice of Acceptance. For the purposes of Contractor's inclusion of "as-built" information, Company will provide Contractor with an electronic version of Company furnished drawings.

11.6 Company reserves the right to review certified material test reports for all materials of construction at any time during field erection. Contractor shall maintain these documents readily available for such review and shall submit all documents to Company on the

completion of the Work.

11.7 Contractor shall maintain at the Work Site up-to-date copies of all drawings, specifications, and other documents and supplementary data, complete with latest revisions thereto. In addition, Contractor shall maintain a continuous record of all field changes, and at the conclusion of the Work, shall incorporate all such changes on the "as built" drawings and other engineering data and shall submit the required number of copies thereof to Company.

11.8 Contractor shall show the Company Contract number and identifying item numbers, if applicable, on all data submitted pursuant to this Article 8.0.

11.9 Where samples are required, they shall be submitted by and at the expense of Contractor. Such submittal shall be made not less than thirty (30) calendar days prior to the time that the materials represented by such samples are needed for incorporation into any Work. Samples shall be subject to review and materials represented by such samples shall not be manufactured, delivered to the Work Site or incorporated into any Work without such review.

11.9.1 Each sample shall bear a label showing Contractor's name, project name, name of the item, manufacturer's name, brand name, model number, supplier's name and reference to the appropriate drawing, technical specification section and paragraph number, all as applicable.

11.9.2 Samples which have been reviewed may, at Company's option, be returned to Contractor for incorporation into the Work.

12.0 CLEAN-UP, SAFETY, WORK RULES AND REGULATIONS

12.1 Contractor shall perform the Work in a safe manner and keep the Work Site in a clean condition, all in accordance with Attachment C, Site Rules and shall comply with all Work rules and regulations set forth in Attachment D, Project Safety & Medical Rules & Regulations.

12.2 Contractor shall submit the Safety Plan and other deliverables required in Attachment D, Project Safety & Medical Rules & Regulations within seven (7) calendar days of award of this Contract.

13.0 QUALITY CONTROL

Contractor shall be responsible for the performance of all inspection and testing activities as specified in Attachment E, Quality Assurance and Control. Contractor shall submit the quality program and inspection procedures required in Attachment E, Quality Assurance and Control within seven (7) calendar days of award of this Contract.

14.0 CONSTRUCTION

Contractor shall perform the Work in accordance with the requirements of this Part I and the documents specifically referenced herein with specific attention to Attachment C, Site Rules.

15.0 KEY PERSONNEL

Contractor shall nominate the following Key Personnel for the Contract.

Executive Sponsor

The role of the Executive Sponsor is to be the Senior Management contact who will become involved and take a proactive approach to the successful execution of the Work. The Executive Sponsor will be aware of the progress of the Work through the major milestones in Exhibit One, Contract Milestone Schedule and will hold at least one meeting each month on the status of the Work.

Contractor's Executive Sponsor will contact Company's Executive Sponsor on any potential problems in Contractor's Organization or in Company's Organization or other major issues that may negatively impact the progress of the Work. Contractor's Executive Sponsor will be available to meet with Company's Executive Sponsor or other Company Project or Construction Management Representative at the worksite to review the status of the Work and the Contract.

Contractor's Executive Sponsor for the contract is _____, who can be reached at (___) _____.

Rolling Stock Project Manager.

The Rolling Stock Project Manager shall be the Contractor's primary contact and shall be responsible for all aspects this scope of work.

Contractor's Rolling Stock Project Manager for the contract is _____, who can be reached at (___) _____.

Rolling Stock Quality Manager

The Rolling Stock Quality Manager shall be the Contractor's primary contract for all Quality related issues for this scope of work.

Contractor's Rolling Stock Quality Manager for the contract is _____, who can be reached at (___) _____.

Rolling Stock Reliability Manager

The Rolling Stock Reliability Manager shall be the Contractor's primary contract for all Reliability, Availability and Maintainability and FMECA related issues for this scope of work.

Contractor's Rolling Stock Reliability Manager for the contract is _____, who can be reached at (___) _____.

The Contractor shall provide a project organization diagram showing the project organization and the personnel assigned to this project

END OF PART I – SCOPE OF WORK